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Master's thesis

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Profile likelihood confidence intervals for derived parameters in dose-response models without reparameterization

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Title and subtitle:	Profile likelihood confidence intervals for derived parameters in dose-response models without reparameterization
Description:	This master thesis introduces a novel method for constructing confidence intervals for derived parameters of dose-response models. Utilizing the profile likelihood method with gradient information offers a reliable and efficient approach without the need for reparameterization. Through simulation studies, the thesis evaluates the performance of this method compared to traditional approaches used in fields like drug development and toxicology.
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1 Introduction

In biomedical and life sciences research, estimating dose-response relationships is fundamental for understanding the effects of various factors on biological systems. Dose-response models play a crucial role in quantifying and characterizing these relationships, identifying optimal dosing, and assessing potential risks associated with different levels of exposure.

Estimating derived parameters, such as effective or benchmark doses, is vital for making informed decisions in areas like drug development, toxicology, and environmental health. Traditionally, confidence intervals for these derived parameters have been obtained through classical asymptotic delta methods or reparameterization techniques, which transform the model parameters to facilitate direct inference with profile likelihood methods. However, the delta method may not be accurate enough in small sample settings, and the profile likelihood method using reparameterization may introduce computational challenges, leading to inaccurate or unreliable results.

To address these limitations, this master thesis investigates an alternative approach to construct confidence intervals for derived dose-response parameters. First, this thesis aims to explore the application of the profile likelihood method enhanced with gradient information to estimate confidence intervals for derived parameters in dose-response models without the need for reparameterization. Doing so aims to provide a reliable and efficient methodology for quantifying uncertainty in derived parameters, enhancing the validity and interpretation of dose-response analyses. Further, the thesis aims to evaluate the performance of the profile likelihood method by conducting simulation studies under various scenarios, considering different dose-response models and sample sizes. The results of these simulations allow us to compare the gradient-based and reparameterization-based profile likelihood methods, highlighting the advantages and limitations of each.

2 Theory

2.1 Likelihood-based methods

Let us first review the classical maximum likelihood theory. Maximum likelihood theory is a powerful statistical methodology providing estimates of the parameters of a statistical model by finding the parameter estimates that are most likely to have generated the observed data. Apart from estimating the parameters, MLE allows us to describe the asymptotic properties of these estimators, test hypotheses about the model parameters, and construct confidence intervals. Its flexibility and solid theoretical foundation has proved to be invaluable for statistical inference.

2.1.1 Review of the maximum likelihood theory

The core terminology for the maximum likelihood estimators (MLE) and their properties is as follows, inspired by Omelka (2023).

Definition 1 (Likelihood function and MLE)**.** *Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ be a random sample of random vectors with a density $f(\mathbf{x}; \boldsymbol{\theta})$ with respect to a σ-finite measure μ, known up to an unknown p-dimensional parameter $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_p)^\top \in \Theta$. Let $\boldsymbol{\theta}_X = (\theta_{X1}, \ldots, \theta_{Xp})^\top$ be the true value of the parameter.*

1. *We define **the likelihood function** as*

$$L_n(\boldsymbol{\theta}) = \prod_{i=1}^{n} f\left(\boldsymbol{X}_i; \boldsymbol{\theta}\right)$$

   *and **the log-likelihood function** as*

$$\ell_n(\boldsymbol{\theta}) = \log L_n(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log f\left(\boldsymbol{X}_i; \boldsymbol{\theta}\right).$$

2. ***The maximum likelihood estimator** of parameter $\boldsymbol{\theta}_X$ is then defined as*

$$\widehat{\boldsymbol{\theta}}_n = \arg\max_{\boldsymbol{\theta} \in \Theta} L_n(\boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta} \in \Theta} \ell_n(\boldsymbol{\theta}).$$

3. ***The score function** of the i-th observation $\boldsymbol{X}_i$ for the parameter $\boldsymbol{\theta}$ is defined as*

$$\mathbf{U}\left(\boldsymbol{X}_i; \boldsymbol{\theta}\right) = \frac{\partial \log f\left(\boldsymbol{X}_i; \boldsymbol{\theta}\right)}{\partial \boldsymbol{\theta}},$$

   *and **the score statistic** as*

$$\mathbf{U}_n(\boldsymbol{\theta}) = \sum_{i=1}^{n} \mathbf{U}\left(\boldsymbol{X}_i; \boldsymbol{\theta}\right) = \sum_{i=1}^{n} \frac{\partial \log f\left(\boldsymbol{X}_i; \boldsymbol{\theta}\right)}{\partial \boldsymbol{\theta}}.$$

4. ***The Fisher information matrix*** *as*

$$I(\boldsymbol{\theta}) = \mathrm{E}_{\boldsymbol{\theta}}\left[\frac{\partial \log f\left(\boldsymbol{X}_i;\boldsymbol{\theta}\right)}{\partial \boldsymbol{\theta}}\frac{\partial \log f\left(\boldsymbol{X}_i;\boldsymbol{\theta}\right)}{\partial \boldsymbol{\theta}^\top}\right].$$

and ***the observed (empirical) information matrix*** *as*

$$I_n(\boldsymbol{\theta}) = -\frac{1}{n}\frac{\partial \mathbf{U}_n(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}^\top} = \frac{1}{n}\sum_{i=1}^{n} I\left(\boldsymbol{X}_i;\boldsymbol{\theta}\right),$$

where

$$I\left(\boldsymbol{X}_i;\boldsymbol{\theta}\right) = -\frac{\partial \mathbf{U}\left(\boldsymbol{X}_i;\boldsymbol{\theta}\right)}{\partial \boldsymbol{\theta}^\top} = -\frac{\partial^2 \log f\left(\boldsymbol{X}_i;\boldsymbol{\theta}\right)}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}.$$

These mathematical objects constitute the base for the MLE theory.

Under some standard regularity conditions (can be found, e.g., in Lehmann and Casella (1998) and will not be discussed any further), the asymptotic behavior of the maximum likelihood estimator can be described as the solution of the likelihood equations $\mathbf{U}_n(\widehat{\boldsymbol{\theta}}_n) = \mathbf{0}_p$.

Theorem 1 (Asymptotic properties of the MLE)**.** *Assume the standard regularity conditions, then*

1.

$$\mathsf{P}(\exists \widehat{\boldsymbol{\theta}}_n : \mathbf{U}_n(\widehat{\boldsymbol{\theta}}_n) = \mathbf{0}_p) \xrightarrow[n\to\infty]{} 1$$

2.

$$\sqrt{n}\left(\widehat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_X\right) \xrightarrow[n\to\infty]{d} \mathsf{N}\left(\mathbf{0}_p, I^{-1}\left(\boldsymbol{\theta}_X\right)\right),$$

where $\widehat{\boldsymbol{\theta}}_n$ is is the solution of the likelihood equations.

Theorem 1 states that under the regularity conditions the MLE exists as a solution of the likelihood equations and further describes the asymptotic normality. The asymptotic normality is especially powerful when constructing confidence intervals. More details can be found in Lehmann and Casella (1998), Section 6.2 – Asymptotic Efficiency.

Now suppose the test of the null hypothesis $H_0 : \boldsymbol{\theta}_X = \boldsymbol{\theta}_0$ against the alternative $H_1 : \boldsymbol{\theta}_X \neq \boldsymbol{\theta}_0$ is of interest. The likelihood ratio test statistic is defined as

$$LR_n = 2\left(\ell_n\left(\widehat{\boldsymbol{\theta}}_n\right) - \ell_n\left(\boldsymbol{\theta}_0\right)\right),$$

and the Wald test statistic as

$$W_n = n\left(\widehat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\right)^\top \widehat{I}_n\left(\widehat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\right),$$

where $\widehat{I}_n$ is an estimate of the Fisher information matrix.

Under some assumptions, generally assumed to be fulfilled when used in practice, it can be shown that both the likelihood ratio and the Wald test statistics converge in distribution to the χ_p^2 distribution.

These results allow the construction of confidence intervals for the parameters of the model. One confidence interval type is based on the Wald test statistic and is called the Wald confidence interval, defined as

$$\left\{\boldsymbol{\theta} \in \Theta : n\left(\widehat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}\right)^\top \widehat{I}_n \left(\widehat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}\right) \leq \chi_p^2(1-\alpha)\right\},$$

where $\chi_p^2(1-\alpha)$ is the $1-\alpha$ quantile of the χ_p^2 distribution. Another often-used confidence interval type is the likelihood ratio test-based confidence interval

$$\left\{\boldsymbol{\theta} \in \Theta : 2\left(\ell_n\left(\widehat{\boldsymbol{\theta}}_n\right) - \ell_n(\boldsymbol{\theta})\right) \leq \chi_p^2(1-\alpha)\right\}.$$

2.1.2 Profile likelihood

The profile likelihood is essential for further developing the framework for calculating confidence intervals. On a high-level overview, the profile likelihood allows dividing the full likelihood by the parameters of interest. That way, only the parameters of interest in the given statistical problem can be analysed and the rest of them can be filtered out – often called *nuisance parameters*.

Definition 2. *Assume the same as in* ***Definition 1****. Further, assume* $\boldsymbol{\theta}$ *can be divided into* $\boldsymbol{\tau}$ *containing the first* q *components and* $\boldsymbol{\psi}$ *containing the remaining* $p-q$ *components, i.e.* $\boldsymbol{\theta} = \left(\boldsymbol{\tau}^\top, \boldsymbol{\psi}^\top\right)^\top = (\theta_1, \ldots, \theta_q, \theta_{q+1}, \ldots, \theta_p)^\top$*. Let the* Θ_ψ *and* Θ_τ *be parameter spaces for* ψ *and* τ *respectively. Rewriting the likelihood function as* $L_n(\boldsymbol{\theta}) = L_n(\boldsymbol{\tau}, \boldsymbol{\psi})$ *(and similarly for the rest of* ***Definition 1****) we define (for the parameter* τ*)*

1. ***The profile likelihood*** *and* ***the profile log-likelihood*** *as*

$$L_n^{(p)}(\boldsymbol{\tau}) = \max_{\psi \in \Theta_\psi} L_n(\boldsymbol{\tau}, \boldsymbol{\psi}) \quad \text{and} \quad \ell_n^{(p)}(\boldsymbol{\tau}) = \log L_n^{(p)}(\boldsymbol{\tau}) = \max_{\psi \in \Theta_\psi} \ell_n(\boldsymbol{\tau}, \boldsymbol{\psi}).$$

and ***the profile maximum likelihood estimate*** *as*

$$\widehat{\tau}_n^{(p)} = \arg\max_{\tau \in \Theta_\tau} \ell_n^{(p)}(\boldsymbol{\tau}).$$

2. ***The profile score statistic*** *as*

$$\mathbf{U}_n^{(p)}(\boldsymbol{\tau}) = \frac{\partial \ell_n^{(p)}(\boldsymbol{\tau})}{\partial \boldsymbol{\tau}}.$$

3. ***The profile Fisher information matrix*** *as*

$$I_n^{(p)}(\boldsymbol{\tau}) = -\frac{1}{n}\frac{\partial \mathbf{U}_n^{(p)}(\boldsymbol{\tau})}{\partial \boldsymbol{\tau}^\top}.$$

Since the uniqueness of the maximum likelihood estimator $\widehat{\theta}_n$ is assumed (Theorem 1), it trivially holds that $\ell_n^{(p)}(\widehat{\tau}_n^{(p)}) = \ell_n(\widehat{\theta}_n)$. Since $\widehat{\tau}_n$ is the first q elements of the θ parameter vector, the $\widehat{\tau}_n^{(p)}$ is the maximum likelihood estimator of these first q elements.

Similarly to the classical MLE described before, analogous test statistics for testing the hypothesis $H_0 : \boldsymbol{\tau}_X = \boldsymbol{\tau}_0$ for the profile likelihood is defined. Namely, the profile Wald test statistic

$$W_n^{(p)} = n\left(\widehat{\boldsymbol{\tau}}_n - \boldsymbol{\tau}_0\right)^\top \widehat{I}_n^{(p)}\left(\widehat{\boldsymbol{\tau}}_n - \boldsymbol{\tau}_0\right),$$

and the profile likelihood ratio based test statistic

$$LR_n^{(p)} = 2\left(\ell_n^{(p)}\left(\widehat{\boldsymbol{\tau}}_n\right) - \ell_n^{(p)}\left(\boldsymbol{\tau}_0\right)\right).$$

Further, the asymptotic distribution can also be shown for the profile test statistics, i.e., both $LR_n^{(p)}$ and $W_n^{(p)}$ converge in distribution to the χ_q^2 distribution, where q is the number of parameters we infer about.

Naturally, both the profile Wald confidence intervals and the profile likelihood confidence intervals are defined as in Section 2.1.1.

3 Dose-response models

Dose-response models are a popular tool used in toxicology and environmental or pharmaceutical sciences to describe the relationship between the dose of a particular chemical agent and the response it causes in a given biological system. Ritz et al. (2016) defines dose-response models as a collection of statistical models having a specific mean structure in common. Dose-response relationships are often modeled via non-linear regression methods – which is the focus of this thesis. The dose-response model generally consists of a model mean function (analogy to a link function in generalized linear regression) and a distributional assumption on the response, given the regressor, referred to as the dose or the concentration.

3.1 Model functions

3.1.1 The log-logistic model

A prominent dose-response model is the log-logistic model, generally described by the equation

$$f(x, \boldsymbol{\theta}) = \mathsf{E}[Y|X, \boldsymbol{\theta}] = c + \frac{d-c}{1+\exp\left(b(\log(x) - \log(e))\right)}, \tag{1}$$

where $\boldsymbol{\theta} = (b, c, d, e)$ are the parameters of the model. A strong advantage of this parameterization is the interpretability of the parameters, namely, parameters c and d are the lower, and the upper asymptote, respectively, and the parameter e is the so-called ED50, i.e., the dose that results in a halfway reduction between the lower and the upper asymptotes of the dose-response curve $f(x)$, and b is a slope parameter. The slope parameter has the opposite sign than would be intuitively expected with positive values corresponding to decreasing curves. We call the model described by equation (1) **the 4-parameter log-logistic model** and refer to it as the **LL.4**.

Further models are described by simplifications of the LL.4. By fixing the lower asymptote to zero, i.e., $c = 0$, we define **the 3-parameter log-logistic model** and refer to it as the **LL.3** and by further fixing the upper asymptote to one, i.e., $d = 1$ we define **the 2-parameter log-logistic model** and refer to it as the **LL.2**.

For later purposes, the gradient of the log-logistic model with respect to the relevant parameters (b, e) is

$$\nabla_{(b,e)} f(x, \boldsymbol{\theta}) = \begin{bmatrix} \frac{\partial f(x,\boldsymbol{\theta})}{\partial b} \\ \frac{\partial f(x,\boldsymbol{\theta})}{\partial e} \end{bmatrix} = \begin{bmatrix} (c-d)\frac{\exp(b(\log(x)-\log(e)))(\log(x)-\log(e))}{(1+\exp(b(\log(x)-\log(e)))^2} \\ (d-c)\frac{b\exp(b(\log(x)-\log(e)))}{e(1+\exp(b(\log(x)-\log(e)))^2} \end{bmatrix}.$$

The reason for only the (b, e) being the relevant parameters will be described in the later sections.

3.1.2 Weibull model

The Weibull model is another prominent dose-response model example. There are two common parameterizations and the so-called Weibull-1 model, (Ritz (2010)), is described as

$$f(x,\boldsymbol{\theta}) = c + (d-c)\exp\left(-\exp\left(b(\log(x)-\log(e))\right)\right). \tag{2}$$

We call this function **the 4-parameter Weibull model** and refer to it as the **W1.4**. The slope parameter b reflects the steepness of the dose-response curve with large values corresponding to steeper curves, similarly to the log-logistic model. The parameter e is the dose where the inflection point of the dose–response curve is located, (Ritz (2010)). This is also true for the log-logistic, however, for the Weibull model e does not correspond to the ED50.

Analogically to the log-logistic models, by fixing the asymptotes c and d we define **the 3-parameter Weibull model** and refer to it as the **W1.3** and **the 2-parameter Weibull model** and refer to it as the **W1.2**.

For further purposes, the gradient of the Weibull model with respect to relevant parameters is

$$\begin{aligned}
\nabla_{(b,e)} f(x,\boldsymbol{\theta}) &= \begin{bmatrix} \frac{\partial f(x,\boldsymbol{\theta})}{\partial b} \\ \frac{\partial f(x,\boldsymbol{\theta})}{\partial e} \end{bmatrix} \\
&= \begin{bmatrix} (c-d)\exp\left(-\exp\left(b(\log(x)-\log(e))\right)\right)\exp\left(b(\log(x)-\log(e))\right)(\log(x)-\log(e)) \\ (d-c)\exp\left(-\exp\left(b(\log(x)-\log(e))\right)\right)\exp\left(b(\log(x)-\log(e))\right)\frac{b}{e} \end{bmatrix}.
\end{aligned}$$

For the comparison, an example of both the Weibull and the log-logistic dose-response curve is seen in Figure 1. Both curves share the same parameters for the asymptotes and the slope. The parameter e differs – for the log-logistic case e = ED50 is set to 50, while for the Weibull model the e is set accordingly, so it holds ED50 = 50. Unlike the log-logistic model, the Weibull model is not symmetric

3.2 Model specification

The full model is defined by adding a distributional assumption on the response variable Y. Cases with the response having a Gaussian, a Poisson, or a binomial distribution are examined.

3.2.1 Continuous response

Firstly, let the response follow the Gaussian distribution. Having both distributional and mean function assumptions, the conditional distribution of the response given the dose can be described as

$$Y|X \sim \mathsf{N}(f(X,\boldsymbol{\theta}); \sigma^2 I_n),$$

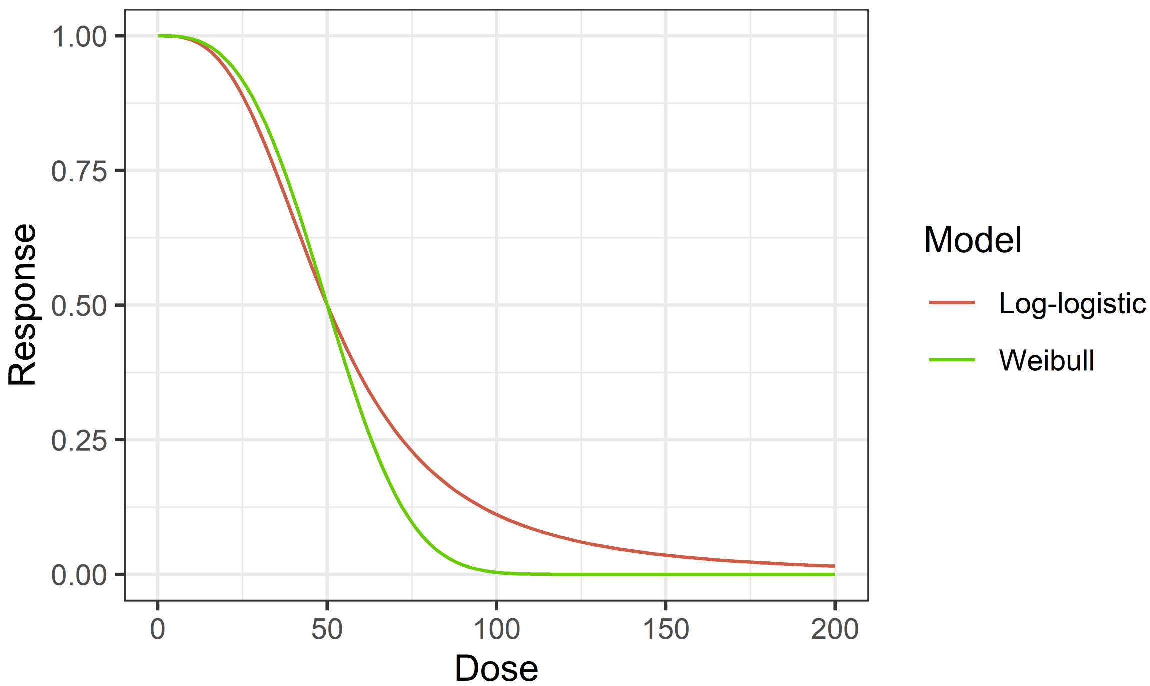


Figure 1: An example of two dose-response curves. The log-logistic one is parameterized as $(b, c, d, e) = (3, 0, 1, 50)$. The Weibull model is parameterized is parameterized as $(b, c, d, e) = (3, 0, 1, 56.5)$. Such parameterizations are chosen so both curves share the same slope and ED50 parameter.

where $f(X, \boldsymbol{\theta})$ is the model function described in Section 3.1. The log-likelihood is then

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log \frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{2\sigma^2} \left(y_i - f(x_i, \boldsymbol{\theta})\right)^2 . \tag{3}$$

As described in Ritz et al. (2016), the maximization of the likelihood for this model reduces to the least squares estimation, i.e., $\arg\max_{\boldsymbol{\theta}\in\Theta} \ell(\boldsymbol{\theta}) = \arg\min_{\boldsymbol{\theta}\in\Theta} \sum_{i=1}^{n} \left(y_i - f(x_i, \boldsymbol{\theta})\right)^2$. This can be seen from the form of the log-likelihood – the least squares expression is embedded in its formula. However, this reduction only applies to the case of continuous response with Gaussian noise.

Having the log-likelihood, the score function is calculated as

$$\begin{aligned}
\mathbf{U}_n(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \ell_n(\boldsymbol{\theta}) &= \sum_{i=1}^{n} -\frac{1}{2\sigma^2} \nabla_{\boldsymbol{\theta}} (y_i - f(x_i, \boldsymbol{\theta}))^2 \\
&= \sum_{i=1}^{n} \frac{1}{\sigma^2} (y_i - f(x_i, \boldsymbol{\theta})) \nabla_{\boldsymbol{\theta}} f(x_i, \boldsymbol{\theta}).
\end{aligned}$$

3.2.2 Poisson response

Compared to the previous Gaussian distribution assumptions, the Poisson distribution models discrete, non-negative values of the response. The regression model is then described as

$$Y|X \sim \mathsf{Po}(f(X, \boldsymbol{\theta})).$$

Poisson regression is generally used when the assumption of homoscedasticity is broken, and a mean-variance relationship exists – typically when the variance of the response grows with growing values of the dose. Poisson regression tries to solve this due to its simple mean-variance relationship, $\mathsf{E}\,\mathsf{Po}(\lambda) = \mathsf{var}\,\mathsf{Po}(\lambda) = \lambda$. The log-likelihood of this model is

$$\ell(\boldsymbol{\theta}) = \log \prod_{i=1}^{n} \mathsf{P}(Y_i = y_i | x_i, \boldsymbol{\theta}) = \sum_{i=1}^{n} -f(x_i, \boldsymbol{\theta}) + y_i \log f(x_i, \boldsymbol{\theta}) - \log y_i!, \tag{4}$$

and the score function is

$$\begin{aligned}
\mathbf{U}_n(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \ell_n(\boldsymbol{\theta}) &= \sum_{i=1}^{n} -\nabla_{\boldsymbol{\theta}} f(x_i, \boldsymbol{\theta}) + y_i \nabla_{\boldsymbol{\theta}} \log f(x_i, \boldsymbol{\theta}) \\
&= \sum_{i=1}^{n} -\nabla_{\boldsymbol{\theta}} f(x_i, \boldsymbol{\theta}) + y_i \frac{1}{f(x_i, \boldsymbol{\theta})} \nabla_{\boldsymbol{\theta}} f(x_i, \boldsymbol{\theta}) \\
&= \sum_{i=1}^{n} \left(\frac{y_i}{f(x_i, \boldsymbol{\theta})} - 1 \right) \nabla_{\boldsymbol{\theta}} f(x_i, \boldsymbol{\theta}).
\end{aligned}$$

3.2.3 Binomial response

In binomial regression, the response Y is assumed to have the binomial conditional distribution given the dose X. The response Y often encodes the number of successes in $\boldsymbol{n}$ trials, where the number of trials $\boldsymbol{n}$ is known. The conditional probability of success p is modeled as a proportion of successes $Y/\boldsymbol{n}$. Modeling the conditional probability naturally leads to the Bernoulli distribution.

The general idea of the binomial regression is modeling the response as a conditionally binomial random variable, i.e.,

$$Y|X \sim \mathsf{Bi}(\boldsymbol{n}, f(X, \boldsymbol{\theta})),$$

where $\boldsymbol{n}$ is a vector with the number of trials. This also implies

$$Y/\boldsymbol{n}|X \sim \mathsf{Bernoulli}(f(X, \boldsymbol{\theta})).$$

The log-likelihood of the model, where y_i is the number of successes in n_i trials, and w_i are the weights (typically set to the number of trials n_i), is

$$\ell(\boldsymbol{\theta}) = \log \prod_{i=1}^{n} \mathsf{P}(Y_i = y_i | x_i, \boldsymbol{\theta}) = \sum_{i=1}^{n} \log \binom{n_i}{y_i} + \frac{y_i}{n_i} w_i \log \left(\frac{f(x_i, \boldsymbol{\theta})}{1 - f(x_i, \boldsymbol{\theta})} \right) + w_i \log(1 - f(x_i, \boldsymbol{\theta})).$$

And finally, the score function,

$$\begin{aligned}
\mathbf{U}_n(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \ell_n(\boldsymbol{\theta}) &= \sum_{i=1}^{n} \frac{y_i}{n_i} w_i \nabla_{\boldsymbol{\theta}} \log \left(\frac{f(x_i, \boldsymbol{\theta})}{1 - f(x_i, \boldsymbol{\theta})} \right) + w_i \nabla_{\boldsymbol{\theta}} \log(1 - f(x_i, \boldsymbol{\theta})) \\
&= \sum_{i=1}^{n} w_i \nabla_{\boldsymbol{\theta}} f(x_i, \boldsymbol{\theta}) \left(\frac{y_i}{f(x_i, \boldsymbol{\theta})} + \frac{y_i}{1 - f(x_i, \boldsymbol{\theta})} - \frac{1}{1 - f(x_i, \boldsymbol{\theta})} \right).
\end{aligned}$$

4 Derived parameters

Derived parameters are of crucial interest in a dose-response analysis as they typically provide a framework for understanding the effects in the model and allow for assessing safe levels of exposure to chemical agents.

Estimates of such parameters are obtained by solving an inverse regression problem, (Ritz et al. (2016)). Inverse regression can be described as follows: First, a certain (non-)linear relationship between two variables is estimated by using pairs of input and output data; thereafter, the value of an unknown input variable is estimated given an observation of the corresponding output variable, (Avenhaus et al. (1980)).

Inferring about these derived parameters is the main interest of this thesis. The following sections describe two prominent examples of the derived parameters, as well as theory and implementation of assessing the uncertainty in their estimation.

4.1 Effective doses

Dixon et al. (2020) describe the effective doses in the following way. When the lower and upper asymptotes of the dose-response curve are not 0 and 1, risk can be described as an extra risk.

Extra risk describes risk as a proportion of the difference between the upper and the lower asymptotes. A 10% extra risk would have a mean response that is 10% from the lower asymptote to the upper asymptote. Extra risk is commonly used when the response is a proportion. In mathematical terms, the $100\alpha\%$ effective dose for a decreasing curve, ED100α, corresponds to the solution of the equation

$$\frac{f(ED100\alpha, \boldsymbol{\theta}) - d}{c - d} = \alpha.$$

Further on a decreasing dose-response curve is assumed. Derivations an increasing variant would be analogous.

The most prominent example is the dose that results in a halfway reduction between the lower and upper limits of the model function f. This parameter is often denoted as ED50 (50% effective dose) or EC50 (50% effective concentration) for continuous responses and LD50 (50% lethal dose) or LC50 (50% lethal concentration) for binomial responses. Note that the ED50 is the parameter e in the log-logistic model.

Calculating the desired effective dose ED100α for the log-logistic model leads to the equation

$$
\begin{aligned}
f(ED100\alpha, \boldsymbol{\theta}) &= (1-\alpha)d + \alpha c \\
c + \frac{d-c}{1+\exp(b(\log(ED100\alpha) - e))} &= (1-\alpha)d + \alpha c \\
&\vdots \\
ED100\alpha &= \exp\left(\frac{1}{b}\log\frac{\alpha}{1-\alpha} + \log e\right).
\end{aligned}
$$

The calculation is analogous for the Weibull model, i.e.,

$$
\begin{aligned}
f(ED100\alpha, \boldsymbol{\theta}) &= (1-\alpha)d + \alpha c \\
c + (d-c)\exp\left(-\exp\left(b(\log(ED100\alpha) - \log(e))\right)\right) &= (1-\alpha)d + \alpha c \\
&\vdots \\
ED100\alpha &= \exp\left(\frac{1}{b}\log\log(1-\alpha)^{-1} + \log e\right).
\end{aligned}
$$

Notice that the formulas for ED100α only include the parameters b, e, since both lower and upper asymptotes parameters c, d are canceled out during the derivation. This is an essential insight for further construction of the confidence intervals for ED100α. The formula can be viewed as a function of the parameters b, e, i.e., ED100$\alpha(b, e)$.

4.2 Benchmark doses

The benchmark dose (BMD) methodology is utilized in toxicology and ecotoxicology to establish a measure of hazard characterization for risk assessment purposes. Suppose the estimated dose or concentration of exposure is lower than the level at which adverse effects are expected. In that case, the risk associated with exposure to the chemical is deemed acceptable, (Jensen et al. (2020)).

The benchmark dose, BMD, is the dose resulting in a pre-specified small change in the response, denoted the BMR, relative to the control group's response level, (Jensen et al. (2020)). Different definitions of BMD exist.

One possible definition, (Jensen et al. (2020)), for continuous and count data is equivalent to the effective doses. The definition of the BMD in terms of extra risk is as follows

$$
\mathrm{BMR} = \frac{f(\mathrm{BMD}, \boldsymbol{\theta}) - f(0, \boldsymbol{\theta})}{f(\infty, \boldsymbol{\theta}) - f(0, \boldsymbol{\theta})}.
$$

In the case of the log-logistic and Weibull model (and generally for dose-response models), the $f(\infty, \boldsymbol{\theta})$ and $f(0, \boldsymbol{\theta})$ are the upper and the lower asymptotes, corresponding to the parameters c

and d, respectively.

The same equivalence with effective doses holds for binomial data in the scenarios where the upper asymptote equals 1. For binomial data the BMD is defined in terms of excess risk (Jensen et al. (2020)) as

$$\text{BMR} = \frac{f(\text{BMD}.\boldsymbol{\theta}) - p_0}{1 - p_0},$$

where the p_0 is the background response level or the background risk and is defined as $p_0 = \lim_{x \to 0} f(x, \boldsymbol{\theta})$. For the log-logistic and the Weibull model for increasing dose-response curve, the limits correspond to the upper asymptote, i.e., $p_0 = d$.

For both continuous and binomial data, there are versions with the risk being defined as an added risk. Added risk describes a multiple of the lower asymptote and does not depend on the upper asymptote. A 10% added risk would have a mean response of 1.1 times the lower asymptote. The added risk is commonly used with continuous responses.

That does not hold in the case of the risk being defined as the added risk –

However, such definitions suffer from certain drawbacks. the parameters corresponding to upper and lower asymptotes would not cancel out, and the functional transformation from parameters to the desired effective dose would have more than two parameters.

Unfortunately, the trick of the upper and the lower asymptote parameters canceling out during the derivation only holds for a small subset of benchmark dose definitions. Described values, together with the uncertainty, can be estimated via the framework for the effective doses.

Further on, this thesis will focus on the effective doses for log-logistic and Weibull models.

4.3 Confidence intervals for effective doses

Computing the estimate of the effective dose based on the MLE of the parameters is straightforward, and such an estimate will be the MLE of the effective dose as well due to the invariance property of the MLE. Finding the confidence interval is more challenging, (Dixon et al. (2020)).

The construction of confidence intervals in Section 2.1 only applies to the model parameters, i.e., the parameters b, c, d, and e ut not to derived parameters such as the effective doses. Further sections describe confidence intervals for effective doses in log-logistic and Weibull models, as in these models, the effective doses are functions of only two parameters.

4.3.1 Delta method confidence intervals

The delta method, (Lehmann and Casella (1998)), is often used for its relative simplicity since it allows derivations of an analytical formula to plug in the MLE estimates. Moreover, the delta method-based confidence intervals are also quite simple to construct for more parameters simultaneously since they consist of a quadratic form and, together with the χ^2 quantile inequality, describe an ellipse.

As previously noted, the ED100α for the log-logistic and Weibull models is a function of only two parameters. The combination of the delta method and maximum likelihood theory yields

$$\sqrt{n}(g(\widehat{b}_n, \widehat{e}_n) - g(b_0, e_0)) \xrightarrow[n\to\infty]{d} \mathsf{N}(0, \nabla g(b_0, e_0)^T I^{-1}(b_0, e_0) \nabla g(e_0, b_0)),$$

where $g(b, e) = ED100\alpha(b, e)$. The delta confidence interval for the $ED100\alpha$ derived parameter is then

$$\left(g(\widehat{b}_n, \widehat{e}_n) - q(1-\alpha)\psi_n,\ g(\widehat{b}_n, \widehat{e}_n) + q(1-\alpha)\psi_n\right),$$

where

$$\psi_n = \frac{\nabla g(\widehat{b}_n, \widehat{e}_n)^T I^{-1}(\widehat{b}_n, \widehat{e}_n) \nabla g(\widehat{b}_n, \widehat{e}_n)}{\sqrt{n}}$$

and $q(1-\alpha)$ is the $1-\alpha$ quantile of the standard normal distribution.

However, delta-type confidence intervals generally suffer from certain drawbacks. Dixon et al. (2020) state that delta (and generally Wald-type) confidence intervals allow the interval to sometimes extend outside of the range of the parameter space. For example, a dose must be greater than or equal to zero, but a delta confidence interval could also include negative values.

4.3.2 Profile likelihood confidence interval with reparameterization

Profile likelihood confidence intervals are based on the likelihood ratio test statistic, and do not extend beyond the parameter space, (Omelka (2023)). This approach has better statistical properties in many situations, as suggested by the Neyman-Pearson lemma, which shows that the likelihood ratio test statistic has the highest power among all test statistics (Anděl (2011), Lehmann and Casella (1998)).

In their paper, Dixon et al. (2020) proposed to calculate profile likelihood confidence intervals by reparameterizing the model such that the desired effective dose is incorporated into the model as a parameter. Profile likelihood confidence interval is then constructed simply using the theory described in Section 2.1.2. Dixon et al. (2020) only provides the reparameterized 4-parameter log-logistic model as

$$f(x, \tilde{\boldsymbol{\theta}}) = c + \frac{d-c}{1 + \exp\left(b(\log(x) - \log(ED100\alpha)) - \log\left(\frac{\alpha}{1-\alpha}\right)\right)}$$

for the increasing dose-response curve (i.e. $b < 0$ and $d > c$), and

$$f(x, \tilde{\boldsymbol{\theta}}) = d - \frac{d-c}{1 + \exp\left(b(\log(x) - \log(ED100\alpha)) - \log\left(\frac{\alpha}{1-\alpha}\right)\right)}$$

for the decreasing dose-response curve (i.e. $b > 0$ and $d > c$).

Nevertheless, reparameterization comes with drawbacks. Belz and Piepho (2015) state that theoretically, substituting individual parameters using reparameterization does not affect the goodness-of-fit and the estimation of the remaining model parameters. In practice, however, some variations may appear. Further, Belz and Piepho (2015) state that reparameterization may lead to a lack of identifiability, and the mathematical reasons might be data-specific and difficult to describe.

4.3.3 Profile likelihood confidence interval without reparameterization

To avoid the reparameterization, a 2-dimensional profile likelihood confidence region for the parameters (b, e) can be constructed and then transformed into the effective dose confidence interval.

Transforming confidence intervals is straightforward as long as the transformation function is bijective (one-to-one), (Smithson (2003)). However, it can be easily shown that the transformed confidence interval has (at least) the same coverage even if the function is only surjective: Let $CI(X)$ be the confidence interval calculated given the data X. Let $g : CI(X) \to T$ be a surjective function. Trivially it holds $\boldsymbol{\theta} \in CI(X) \implies g(\boldsymbol{\theta}) \in g(CI(X))$. Going back to the probability theory, it holds that $\{\omega : \boldsymbol{\theta} \in CI(X)\} \subseteq \{\omega : g(\boldsymbol{\theta}) \in g(CI(X))\}$ and thus $\mathsf{P}(g(\boldsymbol{\theta}) \in g(CI(X))) \geq \mathsf{P}(\boldsymbol{\theta} \in CI(X)) \geq 1 - \alpha$.

Both $W_n^{(p)}$ and $LR_n^{(p)}$ are asymptotically equivalent in distribution, both converging to the χ^2_q distribution. As the Wald-type confidence interval is described as an ellipse through the $W_n^{(p)}$ test statistic, the profile likelihood confidence interval converges to an ellipse as well. However, profile likelihood confidence intervals differ in small-sample behavior. When the sample size is small to moderate, the Wald test is less reliable than the likelihood ratio test and should not be trusted as the distribution of the maximum likelihood estimates may be far from normality, (Agresti (2003), Clayson et al. (2019)).

The confidence interval for ED100α, calculated as a transformation $g(CI(X))$ of 2-dimensional confidence region $CI(X)$ for (b, e), can be written as $g(CI(X)) = (\min(g(CI(X))), \max(g(CI(X)))$, as it is a 1-dimensional interval, and $g(b, e) = ED100\alpha(b, e)$ is a continuous surjective function. Now the problem boils down to finding the minimum and the maximum of the function g on the set $CI(X)$, leading to a bounded optimization problem. Further examining the gradient of the transformation function for the log-logistic model,

$$\nabla ED100\alpha(b, e) = \begin{bmatrix} \frac{\partial ED100\alpha(b,e)}{\partial b} \\ \frac{\partial ED100\alpha(b,e)}{\partial e} \end{bmatrix} = \begin{bmatrix} -\exp\left(\frac{1}{b}\log\frac{1-\alpha}{\alpha} + \log e\right)\frac{1}{b^2}\log\left(\frac{1-\alpha}{\alpha}\right) \\ \exp\left(\frac{1}{b}\log\frac{1-\alpha}{\alpha} + \log e\right)\frac{1}{e} \end{bmatrix} = 0,$$

there is no solution to the equation. Similarly, for the Weibull model, there is no solution to the equation

$$\nabla ED100\alpha(b,e) = \begin{bmatrix} \frac{\partial ED100\alpha(b,e)}{\partial b} \\ \frac{\partial ED100\alpha(b,e)}{\partial e} \end{bmatrix} = \begin{bmatrix} -\exp\left(\frac{1}{b}\log\log\alpha^{-1} + \log e\right)\frac{1}{b^2}\log\log\alpha^{-1}1 \\ \exp\left(\frac{1}{b}\log\log\alpha^{-1} + \log e\right)\frac{1}{e} \end{bmatrix} = 0.$$

A necessary condition for a local extreme is formulated as follows: If a point is an extreme of the function then the derivative at this point is either zero or does not exist. Thus, $ED100\alpha$ has no local extreme. Optimizing a continuous function without local extremes on a bounded set trivially implies that the solution has to lie on the boundary of that set.

This is another important observation, as now we only need to find the boundary of the confidence region for the parameters (b, e).

Calculating multi-dimensional profile likelihood confidence intervals can be challenging as the method does not describe a quadratic form like its Wald-type counterpart. The most basic way to construct a profile confidence interval is to calculate the values of the LRT statistic for all candidate parameter values and construct the interval by selecting all the values that fulfill the inequality condition. However, this can be computationally heavy.

To address this issue, more efficient algorithms have been suggested, such as the method proposed by Venzon and Moolgavkar (1988), which involves solving a system of equations to find the endpoints of confidence intervals for the parameters involved. However, this approach only describes finding 1-dimensional confidence intervals for the model parameters and none of these methods describe the calculation of multi-dimensional confidence regions.

A simple approach to a multi-dimensional confidence region estimation is selecting candidate values on a predefined grid. However, this approach brings several issues, such as the trade-off between the computational time and the precision. Computation exponentially grows with the increasing dimension of the estimated parameter. Finding a viable grid span that is wide enough to cover the actual confidence interval/region but compact enough not to require too much computational time is essential. Therefore, to efficiently calculate the likelihood ratio test-based confidence interval, a more intelligent approach for selecting candidate parameter values needs to be developed.

5 Finding confidence regions

Finding profile likelihood confidence regions, one starts with the MLE estimates $(\hat{b}_n, \hat{e}_n)$ of the parameters b, e. Deciding whether the candidate parameter value (b, e) belongs to the confidence region or not is based on fulfilling the inequality given by the profile LR test

$$2\left(\ell_n^{(p)}(\hat{b}_n, \hat{e}_n) - \ell_n^{(p)}(b, e)\right) \leq \chi_p^2(1-\alpha).$$

Further, ways to choose the candidate values increasingly efficiently are discussed.

5.1 Grid search

Section 4.3.3 mentions that finding the confidence intervals based on the profile likelihood ratio test is complex and often solved by evaluating all the candidate values on a grid.

As noted before, it is not straightforward to set the grid size beforehand, as the size of the search area in the parameter space is unknown. The grid could be expanded during the computation, with the cost of complicated parallelization possibilities. Since only the border of the confidence region is of interest, evaluating all the candidate values inside the confidence region would also be inefficient.

Nevertheless, this method can serve as a valuable validation of different methods and is very simple to implement, not requiring any advanced algorithms or the derivation of gradients of the likelihood.

5.2 Depth-first search

As an alternative to defining the in a rigid way, the grid could be created only around the area where the boundary of the confidence region is thought to be. To ease up the notation, denote

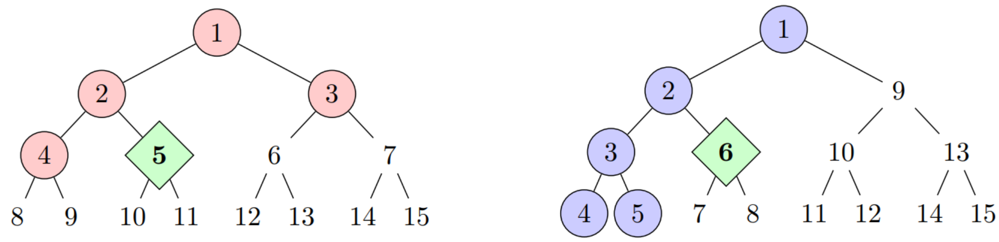
$$\text{test}(b, e) = 2\left(\ell_n^{(p)}(\hat{b}_n, \hat{e}_n) - \ell_n^{(p)}(b, e)\right) - \chi_p^2(1-\alpha),$$

Searching for the boundary of the confidence region can be interpreted as searching for a contour line (more generally a level set) of the $\text{test}(b, e)$ function, specifically, the contour line given by $\text{test}(b, e) = 0$, further referred to as the **boundary**.

Due to the limited precision, let us define the **boundary area** as the set containing a small ϵ-neighborhood ($\epsilon > 0$) around the actual boundary as

$$\{(b, e) : -\epsilon \leq \text{test}(b, e) \leq \epsilon\}.$$

Starting with the MLE $\hat{b}_n, \hat{e}_n$, finding a value on the boundary of the confidence region is of interest. As the MLEs are inside the confidence region, simple incrementation of the value of one



(a) BFS (b) DFS

Figure 2: The difference between BFS (left) and DFS (right) in a complete binary tree where a goal (diamond) is placed in the second position on level 2 (the third row). The numbers indicate traversal order. Circled nodes are explored before the goal is found. Note how BFS and DFS explore different parts of the tree. In bigger trees, this may lead to substantial differences in search performance (Everitt and Hutter (2015)).

of the starting parameters reaches the boundary area, e.g., the candidate values are a sequence $(\hat{b}_n, \hat{e}_n + k\delta)$, $k \in \mathbb{N}$, $\delta > 0$ is the step size, and $\exists l \in \mathbb{N} : -\epsilon < \text{test}(\hat{b}_n, \hat{e}_n + l\delta) < \epsilon$, denoting such a point (b_j, e_j).

Having found the point (b_j, e_j) in the boundary area, we define a grid of points around (b_j, e_j) as $(b_j, e_j+\delta)$, $(b_j+\delta, e_j)$, $(b_j+\delta, e_j+\delta)$, $(b_j, e_j-\delta)$, $(b_j-\delta, e_j)$, $(b_j-\delta, e_j-\delta)$, $(b_j+\delta, e_j-\delta)$, $(b_j-\delta, e_j+\delta)$.

For every point a $-\epsilon < \text{test}(\cdot, \cdot) < \epsilon$ condition is tested and if accepted, such a point again defines a grid of points as described previously.

After expanding the grid around the point in the boundary area, the depth-first search (DFS) algorithm (Cormen and Leiserson (2022)) can be employed. Depth-first search is a tree graph search algorithm, that searches deeper in the graph whenever possible. Depth-first search explores edges out of the most recently discovered vertex that still has unexplored edges. This process continues until a certain goal was accomplished. The goal in the boundary search setting is to calculate a grid covering the whole boundary. The DFS can be implemented recursively.

A visualisation can be seen in Figure 2 (Everitt and Hutter (2015)), where DFS is compared to breadth-first search (BFS), which searches all adjacent vertices first.

Ideally, the whole boundary area should end up being covered by a grid of accepted points. The advantage of the DFS over the BFS algorithm lies in the ability of DFS to finish the procedure without discovering all the acceptable values. The stopping condition can be defined as finding the starting point again, but from a different direction, i.e., after discovering the whole boundary area. A visualisation of this approach can be seen in Figure 3.

Despite the higher efficiency compared to the full, rigid grid, there is still room for improvement. Firstly, each time the grid is expanded around the new viable point, the procedure also looks ”backward” on the boundary area already explored. Further, the grid is rigid, and while it might be too fine for parts of the boundary, increasing the computation cost, it could be too coarse for other parts of the boundary curve, resulting in the inability to find the whole boundary. For its

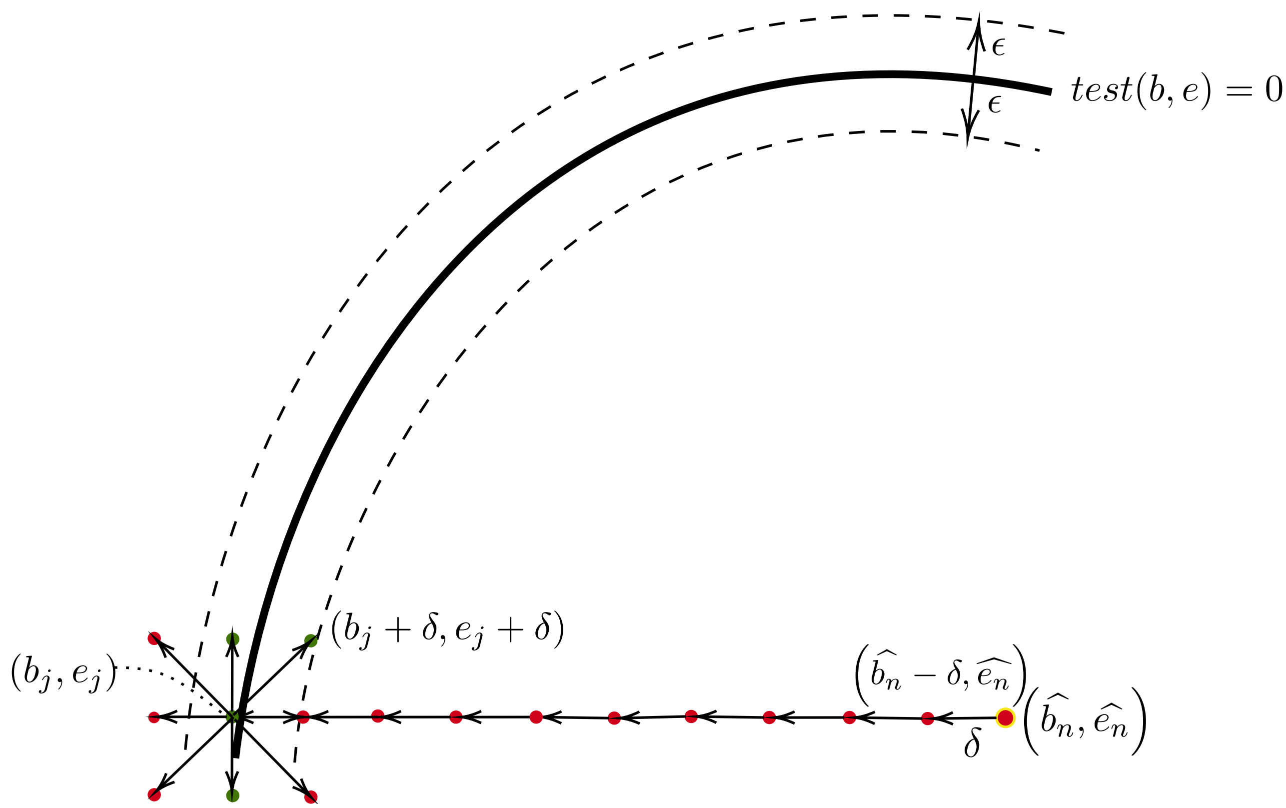


Figure 3: Visualization of the depth-first search approach. The red points are candidate values not belonging to the boundary area, the green points are the points accepted as boundary area points. ϵ is the tolerance bandwidth around the boundary of the confidence region, δ is the step size.

drawbacks, the DFS approach only serves as a middle step in developing a better approach and will not be described further.

5.3 Gradient method

The test(b, e) is a differentiable function as it is a composite function of log-likelihoods, and therefore its gradient can be calculated.

Having the gradient of the test(b, e) function and the MLE estimates $\hat{b}_n, \hat{e}_n$, a point in the boundary area can be found more efficiently than in the case of the DFS. In Section 5.2, one coordinate was incremented until the sequence of candidate points reached the boundary region. The gradient can be utilized instead – similarly to the gradient descent (ascent) methods, (Bishop (2006)). Gradient descent is traditionally used to find a local optimum of a function. However, in this work, the gradient ascent method is used to find the boundary area, i.e., the gradient step looks like

$$(b_{k+1}, e_{k+1}) = (b_k, e_k) + \gamma_k \nabla_{(b,e)} \mathrm{test}(b_k, e_k),$$

where γ_k are certain weights. The MLE $(\hat{b}_n, \hat{e}_n)$ lies in the minimum of the $\mathrm{test}(b, e)$ function. Constructing a sequence $(b_k, e_k)_{k \in \mathbb{N}}$, a certain index exists at which the sequence crosses the boundary, i.e., $\exists j \in \mathbb{N} : \forall k \in \mathbb{N} : k > j$, $\mathrm{test}(b_k, e_k) > 0$.

Contrary to the classical optimization, the optimal value of the function is known – this knowledge is exploited when setting up the weights γ_k. Searching for a (any) point (b, e) such that $\mathrm{test}(b, e) = 0$, the weights can be set to $\gamma_k = \mathrm{test}(b_k, e_k)$. The weights then decrease when approaching the boundary – similarly to having decay schedules in the classical gradient descent, (Bishop (2006)). Setting the weights and normalizing the gradient, the final ascent step procedure is

$$(b_{k+1}, e_{k+1}) = (b_k, e_k) - \lambda \mathrm{test}(b_k, e_k) \frac{\nabla_{(b,e)} \mathrm{test}(b_k, e_k)}{|\nabla_{(b,e)} \mathrm{test}(b_k, e_k)|},$$

where λ is a tuning hyper-parameter. Notice the sign change as $\mathrm{test}(b, e)$ values inside the confidence region are negative. A useful feature of this weighting approach is its sign – when a candidate value gets outside the confidence interval, the sign of the $\mathrm{test}(b, e)$ changes the direction of the ascent back towards the boundary.

The sequence $(b_k, e_k)_{k \in \mathbb{N}}$, is constructed until a $j \in \mathbb{N}$ such that $-\epsilon < \mathrm{test}(b_j, e_j) < \epsilon$ is found. Such a (b_j, e_j) will be the starting point on the boundary. We call this whole procedure `ascent`. Further, the procedure taking the MLE $(\hat{b}_n, \hat{e}_n)$ and returning a point in the boundary area (b_j, e_j), will be called $\mathrm{ascent}(\hat{b}_n, \hat{e}_n) = (b_j, e_j)$.

The key observation is the orthogonality of the gradient and contour lines (level sets) (Strang and Herman (2016)). Section 5.2 creates a grid spanning all directions around each viable point. This section describes a heuristic for only proposing candidate points in a "meaningful" direction. A smooth curve should behave somehow predictably on a small neighborhood around a given point. The "meaningful" direction is estimated based on the gradient. As the gradient is orthogonal to the contour lines, rotating the gradient by 90° should yield the "meaningful" direction. The rotation matrix is defined as

$$R(\alpha) = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}.$$

The essential ingredients to describe the iterative gradient movement alongside the boundary are now collected.

Let (b_j, e_j) be the result of the $\mathrm{ascent}(\hat{b}_n, \hat{e}_n)$. The next candidate point is calculated as

$$(b_c, e_c) = (b_j, e_j) - \frac{\lambda}{|\nabla_{(b,e)} \mathrm{test}(b_k, e_k)|} R(\alpha) \nabla_{(b,e)} \mathrm{test}(b_k, e_k).$$

This procedure on the previously found boundary point (b_j, e_j) will be called `move` and denoted

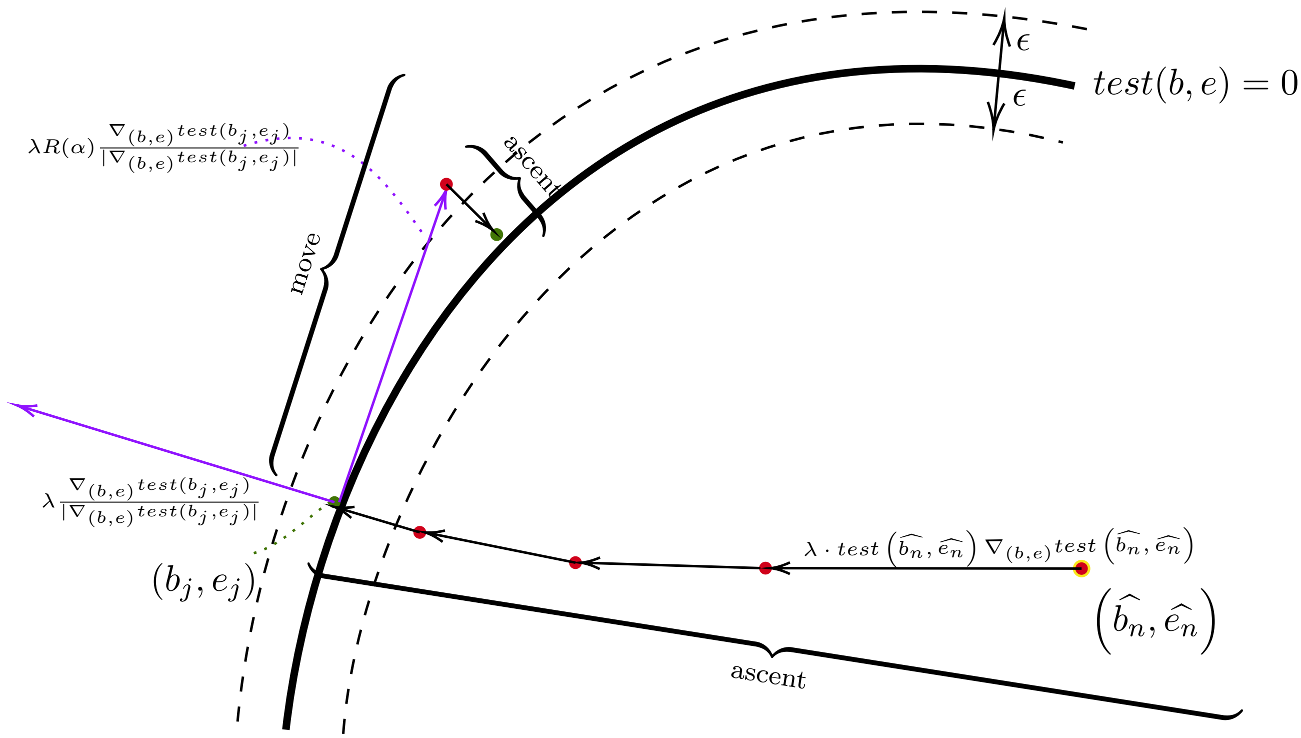


Figure 4: Visualization of Algorithm 1. The red points are candidate values not belonging to the boundary area, the green points are the points accepted as boundary area points. ϵ is the tolerance bandwidth around the boundary of the confidence region, δ is the step size. Firstly, the method finds a point (b_j, e_j) on the boundary with the gradient ascent procedure. From this point, a rotated gradient step is made to find a "meaningful" candidate for the next boundary point. This candidate lies outside the tolerance band, thus another gradient ascent procedure is called to converge to the boundary area again. The rotated gradient step is repeated until the whole boundary is discovered.

$\text{move}(b_j, e_j) = (b_c, e_c)$.

The condition $-\epsilon < \text{test}(b_c, e_c) < \epsilon$ is verified and if not fulfilled, the procedure `ascent` is called and the output is again denoted as $(b_j, e_j) = \text{ascent}(b_c, e_c)$. The condition $-\epsilon < \text{test}(b_c, e_c) < \epsilon$ is not always fulfilled, as a non-linear function is being approximated by a linear one.

These steps concise the gradient-based movement heuristic to choose candidate values in a "smart" way. The procedures `move` and `ascent` are repeated until a sufficient amount of points on the whole boundary is found. The sufficient amount of points on the boundary has to be assessed by the user from a diagnostic plot. Diagnostic plots are described in later sections. The algorithm is described in pseudo-code in Algorithm 1 and visualized in Figure 4.

Algorithm 1 Gradient-based boundary search

Input: Dose-response model, $\epsilon > 0$, $\delta > 0$, $\lambda > 0$
Output: Set of boundary points

Function definition

```
function ASCENT(candidate)
    while not −ε < test(candidate) < ε do
        gradient ← λ ∗ test(candidate) ∗ ∇test(candidate) / |∇test(candidate)|
        candidate ← candidate − gradient
    end while
    return candidate                                        ▷ Boundary point
end function
function ROTATE(α)
    rotation ← ( cos(α)  −sin(α) ; sin(α)  cos(α) ).
    return totation
end function
function MOVE(candidate)
    gradient ← λ ∗ test(candidate) ∗ rotate(α) ∗ ∇test(candidate) / |∇test(candidate)|
    candidate ← candidate − gradient
    return candidate
end function
```

$$\text{rotation} \leftarrow \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}.$$

Implementation

```
dataframe ← 0
beginning ← ascent(MLE of the dose-response model)
candidate ← ascent(move(beginning))
dataframe ← append(dataframe, beginning)
while |beginning - candidate| > δ do
    candidate ← ascent(move(candidate))
    dataframe ← append(dataframe, beginning)
end while
return dataframe
```

6 Implementation

The gradient-based search algorithm was implemented on top of the `drc` R package by Ritz et al. (2016). The `drc` is a robust package providing a comprehensive set of functions and procedures for dose-response modeling. The package provides an implementation for calculating Wald confidence intervals for the model parameters and delta method-based confidence intervals for custom effective doses. However, as discussed before in Section 4.3.3, these methods may not be reliable in small sample size settings, (Agresti (2003)).

The gradient-based profile confidence interval method developed above is implemented as a function `EDx()`. The `EDx()` requires

- `object`: the fitted `drm` model; no default,
- `perc`: the vector of the effective dose percentages, e.g., `c(10, 20)`, for $ED10$ and $ED20$; the default is `perc=c(50)`,
- `level`: the confidence interval coverage; the default is `level = 0.95`
- `lambda`: the tuning parameter λ; the default is `lambda=NULL`,
- `epsilon`: the precision parameter; the default is `epsilon=1e-2`,
- `maxiter`: the maximum number of points on the boundary to be found unless the stopping condition is met; the default is `maxiter=2000`,

and returns a table with ED estimates, corresponding profile likelihood confidence intervals, and a diagnostics plot.

The δ parameter described in Algorithm 1 is determined automatically based on the data.

A heuristic approach is implemented to determine the tuning parameter λ based on the input data. This heuristic is applied when the `lambda` is left unspecified (i.e. `lambda=NULL`). However, the heuristic is not generally robust, and while testing showed sufficient robustness for the LL.4 model with a continuous response, the performance is not optimal for binomial and Poisson models.

In case of failed convergence, it is advised to experiment with the `lambda` parameter and assess the quality of the result visually based on the provided diagnostic plot. Generally, it is recommended to start with `lambda` parameters between 0.01 and 0.05. This recommendation is relevant for all types of models and scales of the data, as both the data-specific δ parameter and a particular internal rescaling works towards standardizing this tuning parameter. However, modeling can still be very sensitive to the value of the λ parameter.

The provided implementation covers all types of log-logistic and Weibull models described in Section 3.1, and all of them can be freely combined with continuous, Poisson, and binomial responses in the way described by Ritz et al. (2016). However, some combinations make more sense than others, and a certain degree of caution is advised. Implemented methods are in Table 1.

Typically, the two-parameter models LL.2 and W1.2 are best suited for the proportional response, i.e., the conditional probability in the binomial regression. Using the basic two-parameter models for continuous data should be well justified prior to the modeling. For example, combining count data (often modeled with Poisson regression) with two-parameter models with upper and lower asymptote one and zero would usually result in a completely wrong and not biologically plausible model.

Often, the 3-parameter models LL.3 and W1.3 can be useful when one has a biological intuition about the lower limit being 0. This is often a viable assumption with count data.

It is useful to note that models with more parameters typically carry more uncertainty, resulting in increased computational times, as the profiling has to be done over multiple nuisance parameters.

Data type:	Continuous, Binomial, Poisson
Model:	LL.2, LL.3, LL.4, W1.2, W1.3, W1.4

Table 1: Models for which the gradient-based profile likelihood confidence intervals are implemented.

The implementation is quite simple to use. An easy example would be an LL.4 model fitted on the ryegrass data. The example of ryegrass data is further discussed in the latter sections. The following code fits a model with the `drc` package and calculates the profile likelihood confidence intervals for ED1, ED5 and ED10

```
> m1 = drm(rootl ~ conc, data = ryegrass, fct = LL.4())
> EDx(m1, c(1,5,10))
      Estimate      2.5 %    97.5 %
ED1  0.6550179 0.3999425 1.014160
ED5  1.1393009 0.8235958 1.531009
ED10 1.4637057 1.1386611 1.849677
```

A table in the terminal and a diagnostic plot (can be seen in later section in Figure 6) are produced.

Further, the calculated confidence interval is of the newly defined class ”profileCI,” and the object contains information gathered during the computation – all of the candidate values together with a plot (for possible further diagnostic purposes), boundary values of the confidence region (can be useful for custom derived parameters) and measured run time.

The code is accessible at GitHub, and can be seen in Appendix A.

7 Practical results

7.1 Ryegrass dataset

7.1.1 Dose-response analysis

The example data are a part of the study by Inderjit et al. (2002) to investigate the joint action of phenolic acids on the root growth inhibition of the perennial ryegrass (Lolium perenne L). The variable `conc` is the concentration of the ferulic acid in mM, and `rootl` is the root length of perennial ryegrass measured in cm.

Modeling the dose-response relationship with the 4-parameter log-logistic function and normal errors via the `drc` package and calculating the Wald confidence intervals for the parameter yields the results in Table 2. Figure 5 shows the data and the fitter dose-response model.

	Estimate	2.5%	97.5%
b : slope parameter	2.982	2.012	3.952
c : lower asymptote	0.481	0.039	0.924
d : upper asymptote	7.793	7.399	8.187
e : ED50	3.058	2.671	3.445

Table 2: The estimates and the Wald confidence intervals for the parameters of the log-logistic model with normal errors fitted to the ryegrass data.

Further, the effective doses ED1, ED5, ED10, i.e., the doses resulting in $1\%, 5\%, 10\%$ decrease in the root length respectively, is of interest. The effective dose `R` function provides the estimates together with the delta method confidence intervals. The results can be seen in Table 3 together with the results of the gradient-based profile likelihood method. The diagnostic plot in Figure 6 (confidence region used to calculate the confidence interval for the effective doses) does not indicate any problems.

The delta and the gradient-based profile likelihood confidence intervals differ slightly (as expected) in this case. The similarity could be a consequence of a good fit of the model and sufficient data.

	Estimate	Delta		Profile	
		2.5%	97.5%	2.5%	97.5%
ED1	0.655	0.320	0.990	0.399	1.015
ED5	1.139	0.751	1.526	0.823	1.532
ED10	1.464	1.074	1.853	1.138	1.850

Table 3: The estimates, the delta, and the gradient-based profile confidence intervals for the effective doses, estimated for the ryegrass dataset.

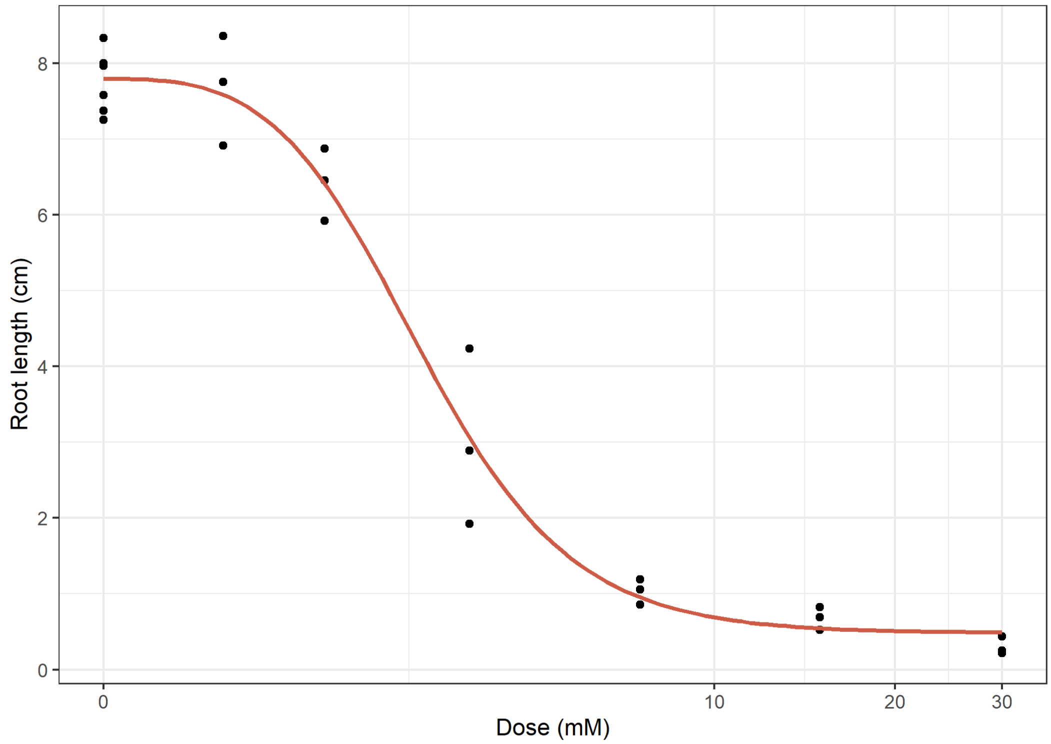


Figure 5: Raw data from the ryegrass dataset with the fitted dose-response model.

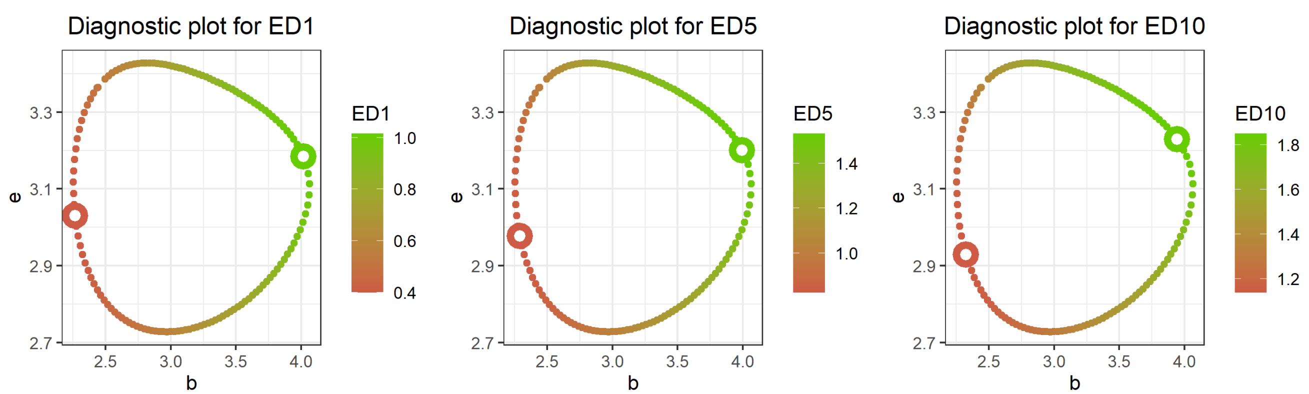


Figure 6: Confidence regions with color-coded values of the given effective dose per estimated point on the boundary and the minimum and the maximum emphasized for visual diagnostics for the ryegrass data. The diagnostic plot is used for a visual assessment of the method convergence and the sufficient coverage.

7.1.2 Model misspecification

Staying with the ryegrass dataset, let us explore a case of a model misspecification. Instead of modeling the response as a continuous variable with normal errors, a Poisson distributed response is assumed instead. The coefficients of the log-logistic model with Poisson distributed errors fitted on the ryegrass dataset are in Table 4. The confidence intervals are wider than in the previous case with normal errors. The confidence interval for the parameter b is noteworthy, as it covers both positive and negative values – thus not rejecting the hypothesis of the slope being 0.

The misspecification of the model and wide confidence intervals become apparent when looking at the estimated confidence region for the parameters (b, e) in Figure 7. Although theoretically converging to the elliptic shape, due to the small sample size and the model misspecification, the shape is deformed (not even convex), and the algorithm failed to find the whole border and stopped after 2000 iterations. The failed convergence comes with an implemented warning message. Finding the whole border could be achieved by increasing the limit on the maximum of iterations or increasing the step size. However, for this more or less hypothetical example, it is beneficial to show how even a failed calculation can still be helpful.

The upper and lower limits of the confidence interval for the effective dose are estimated as the maximum and minimum points on the boundary, respectively. Even though the boundary is not fully found, the diagnostic plot (Figure 7) can assess whether at least one of the confidence interval limits are useful.

The lower confidence interval limit might still be relevant, as the minimum ED value of the confidence region were captured. However, visually it seems that the upper limit was not entirely captured and should not be fully trusted. However, capturing only the lower limit of the confidence interval can be useful as well. In Jensen et al. (2019) the need for the lower limit of benchmark doses (BMDL) is described. This can be useful, e.g., when trying to find the lower limit of a safe dose in risk assessment setting. Thus the gradient-based profile likelihood confidence interval method can still be useful in the benchmark dose setting (where the gradient-based method is applicable).

The drawbacks of the delta method-based confidence intervals are apparent in Table 5 where the lower limits of the confidence intervals for effective doses are negative, i.e., outside the parameter space.

7.2 An illustrative example

This section describes the behavior of the gradient-based profile confidence interval search algorithm, compared to the implementation of profile confidence intervals with reparameterization by Dixon et al. (2020). Due to the limitations of the implemented reparameterization, only the case of the LL.4 and the continuous response is explored. The behavior of the gradient-based algorithm differs for different models and is often very dataset-specific.

	Estimate	2.5%	97.5%
b : slope parameter	2.634	-0.0282	5.297
c : lower asymptote	0.407	-0.313	1.127
d : upper asymptote	7.864	5.903	9.823
e : ED50	3.045	1.632	4.457

Table 4: The estimates and the Wald confidence intervals for the parameters of the log-logistic model under a model misspecification for the ryegrass dataset.

	Estimate	Delta		Profile	
		2.5%	97.5%	2.5%	97.5%
ED1	0.532	-0.519	1.584	0.032	3.636*
ED5	0.995	-0.364	2.356	0.152	3.682*
ED10	1.322	-0.128	2.774	0.302	3.703*

Table 5: The estimates, the delta, and the profile confidence intervals for the effective doses under a model misspecification for the ryegrass dataset. The values with a star are biased due to a failed convergence.

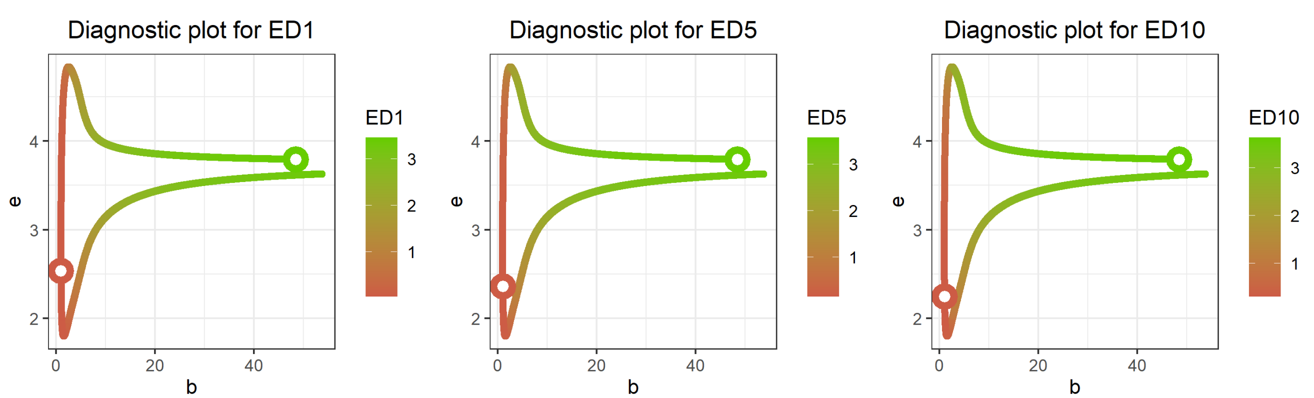


Figure 7: Confidence regions with color-coded values of the given effective dose per estimated point on the boundary and the minimum and the maximum emphasized for visual diagnostics for the ryegrass data. The diagnostic plot is used for a visual assessment of the method convergence and the sufficient coverage. The results are under the model misspecification.

We investigate how the shape of the confidence region and the consequent convergence behaves based on the underlying data. The focus is on the cases when the full span of the dose-response curve is not observed.

In previous sections, claims were made that the profile likelihood-based confidence intervals are valuable in the small sample settings. To explore the small sample behavior, a scenario with seven doses (0, 4.03, 5.31, 7.44, 11.46, 18.16, and 20) in two replicates based on simulated data are considered. These are randomly drawn numbers between 0 and 20, with the endpoints added. The response is sampled from a 4-parameter logistic dose-response model with varying parameter values to showcase the robustness of the method, and confidence intervals for ED1, ED5, and ED10 are calculated by the gradient-based method, the reparameterization, and the classical delta method. Three datasets will be simulated from three models. The model parameters remain fixed with the exception of the ED50 parameter e, which is varied. Further, in model f_1 we set $e = 5$, in f_2 we set $e = 10$ and in f_3 we set $e = 15$ and add the standard Gaussian noise. The generated responses with fitted dose-response curves can be seen in Figure 8.

First, let the generative model f_1 be

$$f_1 : Y|X \sim \mathsf{N}(f(X, 3, 0, 10, 5); I_n).$$

The estimated confidence intervals can be seen in Table 6. The two profile likelihood methods seem to yield similar results (as they should). This also serves as a validation of the developed methodology. Delta-based confidence intervals suffer from the same drawbacks as previously – extending outside the parameter space. However, all of them cover the true value of the parameter. The diagnostic plot in Figure 9a suggests that the gradient-based method succeeded in finding the whole confidence region boundary.

	True	Estimate	Gradient-based		Reparameterization		Delta	
			2.5%	97.5%	2.5%	97.5%	2.5%	97.5%
ED1	1.080	0.890	0.353	1.569	0.296	1.532	-0.428	2.208
ED5	1.873	1.769	1.014	2.562	0.857	2.503	0.113	3.426
ED10	2.403	2.415	1.620	3.214	1.375	3.117	0.721	4.109

Table 6: The true values of the parameter, estimates, and different types of confidence intervals for effective doses for data generated from the model f_1.

Further, let the generative model be

$$f_2 : Y|X \sim \mathsf{N}(f(X, 3, 0, 10, 10); I_n).$$

The estimated confidence intervals in Table 7 indicate more issues than encountered in the case of the model f_1. The implementation of the reparameterized model failed to find the confidence interval. The diagnostic plot in Figure 9b indicates the success of the boundary search, even with the actual shape of the confidence region being further from the ellipse (as assumed in the delta method). Also, both the gradient-based and delta method confidence intervals cover the true value of the parameter.

	True	Estimate	Gradient-based		Reparameterization		Delta	
			2.5%	97.5%	2.5%	97.5%	2.5%	97.5%
ED1	2.161	2.896	1.670	4.842	NA	NA	0.060	5.732
ED5	3.747	4.196	2.998	5.657	NA	NA	1.632	6.761
ED10	4.807	4.963	3.869	6.075	NA	NA	2.672	7.255

Table 7: The true values of the parameter, estimates, and different types of confidence intervals for effective doses for data generated from the model f_2.

Further, let the generative model be

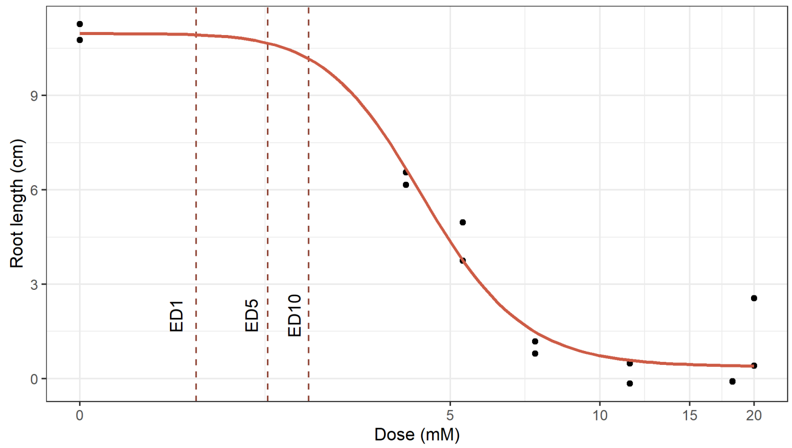
$$f_3 : Y|X \sim \mathsf{N}(f(X, 3, 0, 10, 15); I_n).$$

The estimated confidence intervals in Table 8 indicate issues with the reparameterized model as in the previous case. The gradient-based method reached the maximum iteration limit and failed to find the whole boundary (Figure 9c) of the strongly deformed and non-convex confidence region. However, it seems that the lower limit of the confidence interval is still correct, as the part of the curve responsible for the values close to the lower limit was well described. On the other hand, the upper limit can only be viewed as a lower estimate of the upper limit, as the part of the curve responsible for high values of ED was not fully found.

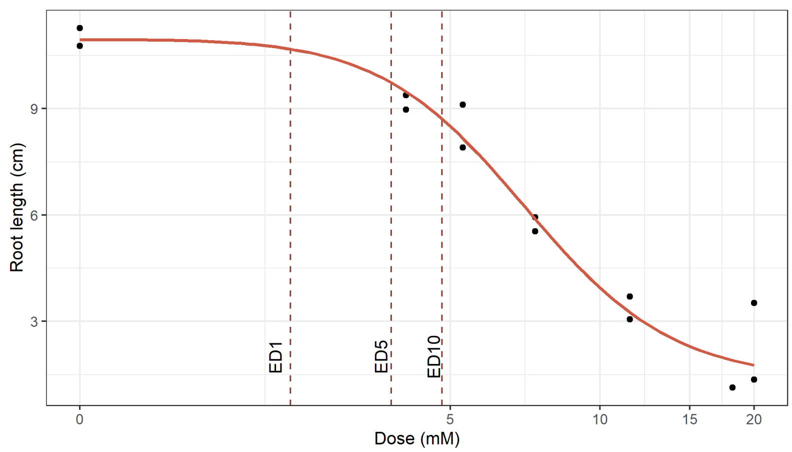
	True	Estimate	Gradient-based		Reparameterization		Delta	
			2.5%	97.5%	2.5%	97.5%	2.5%	97.5%
ED1	3.242	5.701	2.315	11.110	NA	NA	-3.630	15.032
ED5	5.621	7.167	4.278	11.244	NA	NA	-0.522	14.856
ED10	7.211	7.948	5.488	11.305	NA	NA	1.400	14.497

Table 8: The true values of the parameter, estimates, and different types of confidence intervals for effective doses for data generated from the model f_3.

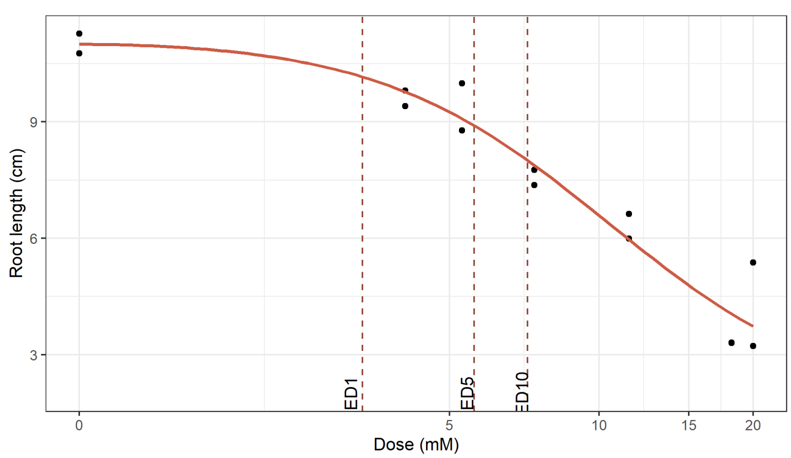
An interesting trend can be observed from the diagnostic plots. Loosely speaking, the deformation of the confidence region in the LL.4 with continuous response can be viewed as a function of "how close the ED50 is towards the high end of doses observed". In the last case, where the ED50 is 15, and the doses span from 0 to 20, the right side of the curve is not well described and leaves a lot of space for uncertainty (especially regarding the parameter responsible for the lower asymptote). The strong deformation can be partially fixed using a different model function, e.g.,



(a) f_1

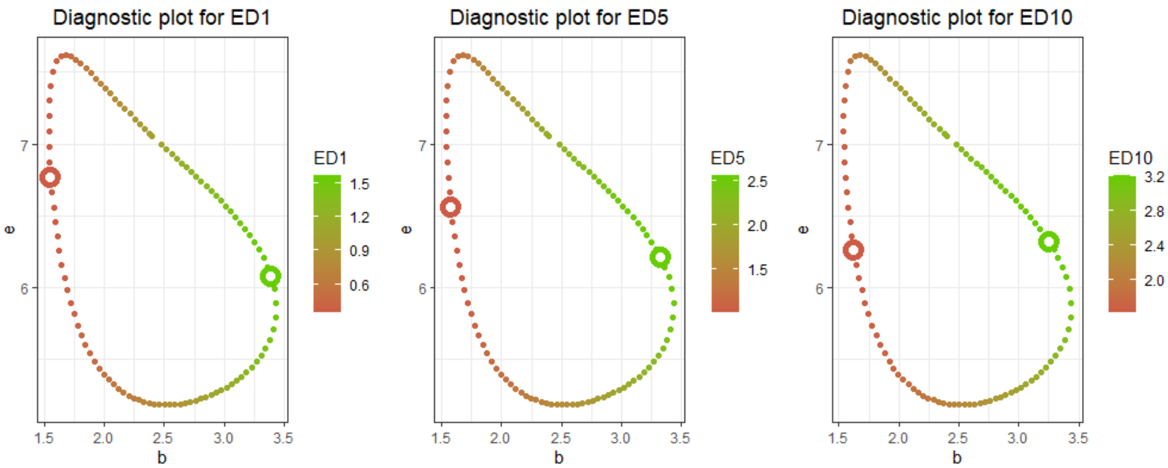


(b) f_2

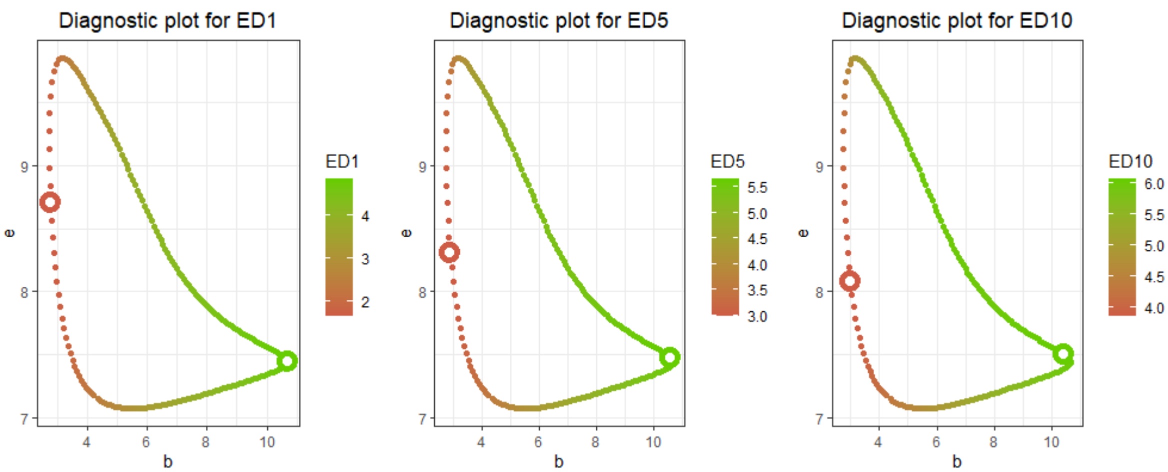


(c) f_3

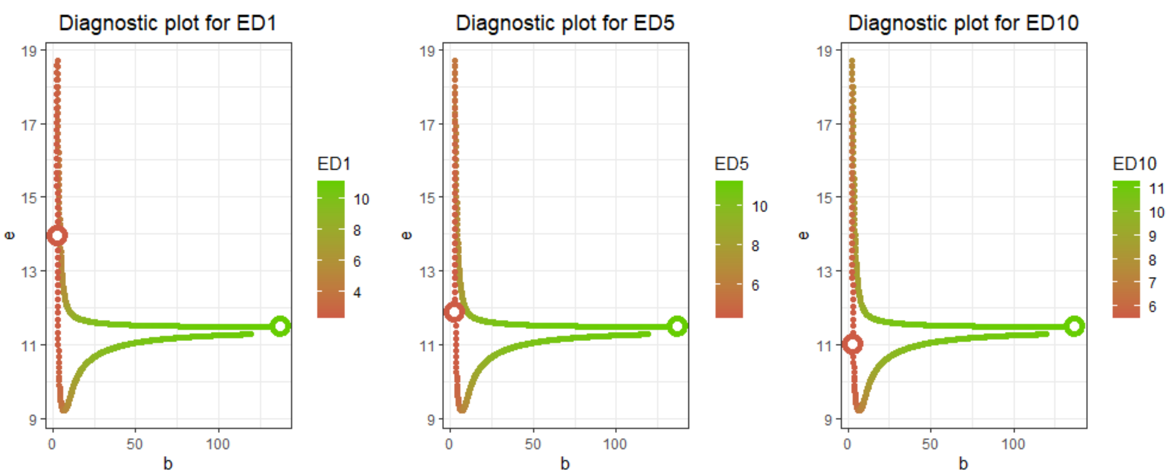
Figure 8: Plots showing the generated data together with the dose response curve for the illustrative example. The true effective dose values are highlighted in each figure.



(a) Diagnostic plot for the model f_1.



(b) Diagnostic plot for the model f_2.



(c) Diagnostic plot for the model f_3.

Figure 9: Diagnostic plots for the three described models on simulated data.

the LL.3, where the lower asymptote is fixed to be 0 and removes some of the uncertainty in the model.

On a higher level, the visual assessment of the confidence region can serve as an additional method to evaluate the model's goodness of fit with elliptic shapes indicating correct modeling. However, more research would have to be done to make any strong claims.

This small example outlines some of the challenges of using the profile likelihood methods. The reparameterization implemented by Dixon et al. (2020) suffers from the lack of robustness, and even this example had to be cherry-picked to illustrate it working on some particular data. The lack of feasible results also makes it more challenging to assess the quality of the new gradient-based method.

On the other hand, the gradient-based method failed to find the entire boundary in the last case of the deformed confidence region. That can be tuned by increasing the maximum iteration limit; however, that does not redeem all the issues. Even with the higher maximum iteration limit, the surface of the $\text{test}(b, e)$ function gets very flat in the cases of the confidence region being too deformed and renders it difficult to find the boundary in some cases. Cases with deformed confidence regions also come with highly inflated computation times. While in the case of f_1, the confidence region can be found in a matter of seconds, the case f_3 is the order of minutes.

7.3 Simulation study

The simulation study attempts to calculate the empirical coverage probability of the gradient-based profile likelihood confidence intervals. The original goal was to compare the new method (mainly) with the reparameterization technique. However, due to the lack of the robustness, only the comparison to the delta method was done.

The simulation study aims to explore settings in multiple directions, i.e., varying the sample size, the model and the data type.

The data is sampled similarly as in Section 7.2. The doses are the same as in Section 7.2, but with different numbers of replications in each experiment. The response is sampled as

$$Y|X \sim \mathsf{N}(f(X, \mathrm{Par}), \sigma^2 I_n), \tag{5}$$

for the continuous case, and

$$Y|X \sim \mathsf{Po}(f(X, \mathrm{Par})), \tag{6}$$

for the Poisson case.

The variance σ^2 in the continuous case is set to 1. Response is generated from models (5) and (6) with varying parameters.

The empirical coverage is the proportion of the 95% confidence intervals covering the true value of the parameter. 200 responses are generated for each model given by the parameter setup in

Table 9. Theoretically, the results of the empirical coverage should converge to 0.95, as that is the significance level for which the confidence intervals are constructed. The gradient-based method was used with the default maximum number of iterations (2000) and λ was manually set to 0.02.

Model	Data	Par	Rep	Gradient-based				Delta		
				ED0.1	ED1	ED10	Conv	ED0.1	ED1	ED10
LL.4	Cont	(3, 0, 10, 5)	2	0.90	0.90	0.89	0.78	0.90	0.92	0.95
LL.4	Cont	(3, 0, 10, 5)	3	0.96	0.97	0.97	0.89	0.96	0.98	0.98
LL.4	Cont	(3, 0, 10, 5)	5	0.94	0.96	0.97	0.97	0.92	0.95	0.97
W1.4	Cont	(3, 0, 10, 5)	2	0.91	0.92	0.93	0.91	0.91	0.93	0.96
W1.4	Cont	(3, 0, 10, 5)	3	0.94	0.94	0.94	0.95	0.93	0.94	0.95
W1.4	Cont	(3, 0, 10, 5)	5	0.95	0.95	0.96	0.99	0.92	0.95	0.97
LL.4	Po	(3, 0, 10, 5)	2	0.81	0.78	0.78	0.07	NA	NA	NA
LL.4	Po	(3, 0, 10, 5)	3	0.87	0.87	0.84	0.18	0.85	0.91	0.89
LL.4	Po	(3, 0, 10, 5)	5	0.93	0.90	0.85	0.23	0.85	0.88	0.90
W1.4	Po	(3, 0, 10, 5)	2	0.71	0.70	0.67	0.09	0.75	0.76	0.78
W1.4	Po	(3, 0, 10, 5)	3	0.76	0.77	0.76	0.17	0.82	0.81	0.82
W1.4	Po	(3, 0, 10, 5)	5	0.83	0.84	0.83	0.21	0.84	0.84	0.83

Table 9: Table comparing the empirical coverage probability across different settings of the model, data type, parameters and repetitions and percentage of trials that converged (for the gradient-based profile likelihood method).

The results in Table 9 reveal that in the continuous case, the proportion of trials that converged is significantly higher than in the Poisson case. That might suggest, that the confidence regions for parameters (b, e) in Poisson models are more deformed in this configuration than in the continuous case. Further, the low convergence rates in the Poisson case might stem from wrong hyperparameter settings, i.e., low maximum numbers of iterations and possibly a low tuning parameters λ, responsible for adjusting the step size.

The results suggest decent coverage for log-logistic and Weibull models with continuous response. The empirical coverages for Poisson configuration look worse, namely for the very small sample settings with 2 and 3 repetitions.

Overall, the empirical coverage seems to grow with the proportion of experiments that converged. Further experiments with increasing the maximum number of iterations in the Poisson case resulted in improvement in the empirical coverage, but greatly increased computation times.

The binomial case was not explored further in the simulation study as it proved to be difficult to set the λ tuning parameter uniformly throughout the experiment.

8 Discussion

The simulation study in Section 7.3 revealed an acceptable robustness for the case of a continuous response variable. Gradient-based profile likelihood confidence intervals proved to be more challenging to calculate in the Poisson case and would probably require more tuning by hand. There seems to be a relationship between the empirical coverage and the proportion of experiments that converged, suggesting that increasing the quality of convergence might increase the empirical coverage. However, further research would be needed.

The main limitation of the gradient-based method is the assumption that the effective doses are only a function of 2 parameters of the model and that the rest of the parameters cancel out during the derivation of the ED. However, that is only the case when working with a few particular models, such as the log-logistic or Weibull type 1 models with up to 4 parameters.

In modern dose-response analysis, more complicated models capture phenomena such as hormesis, (Belz and Piepho (2015)). Such models include, e.g., the Cedergreen-Ritz-Streibig model

$$f(x, \boldsymbol{\theta}) = c + \frac{d - c + \alpha x^{\beta}}{1 + \exp(b(\log x - \log e))}$$

proposed by Cedergreen et al. (2005). However, calculating effective doses for such a model would be more complicated as the confidence region would have to be constructed for more than two parameters. The issue is essentially the same as in the case of BMD definitions based on the added risk. However, the Cedergreen-Ritz-Streibig model might be even more challenging due to even higher number of parameters.

To accommodate for the higher number of parameters, the current framework would have to be further developed. For the case of a 3-dimensional confidence region, the estimation of the whole boundary (a surface in this case) could be divided by slicing the 3-dimensional object and applying the already developed 2-dimensional confidence region boundary estimation. The 3-dimensional confidence region could be sliced by either fixing (and incrementing) one parameter value, or radially, defining planes intersecting the MLE estimates. Similar techniques might work in higher dimensions as well, but with exponentially increasing computational time. The visual diagnostics might not be feasible in higher dimensions of the confidence region. Other way of assessing the quality of the convergence would likely have to be developed.

The other limitation stems from the transformation used to calculate derived parameters from the model parameters. For example, the effective dose transformation for log-logistic and Weibull models is shown to have no local optima in Section 4.3.3. The same might not be true for more complex derived parameters, such as some of the most common benchmark doses, e.g., the BMD defined by the hybrid definition (Jensen et al. (2020)). In such a case, not only the boundary of the confidence region, but also its interior would have to be explored. Exploring the interior of the confidence region would require developing further techniques to search through the interior values and burden the researchers with significantly inflated computation times.

The thesis's original goal was to implement the "out of the box" solution – a versatile function capable of calculating profile likelihood confidence intervals for derived parameters of different models and data types. Unfortunately, automatically tuning the λ parameter, responsible for altering the step size, proved more difficult than anticipated, as the values of λ useful in practice are heavily dependent on the data and the model used. A heuristic for determining "a good" value of λ was implemented; however, its robustness is not always sufficient, and a certain degree of manual tuning might often be required. The heuristic starts with the $\lambda = 1$ and tries to estimate the boundary. If λ is too high, the estimation results in a crash of the profiling procedure. The parameter λ is being decreased until no crash occurs.

The computation times of the gradient-based method, especially for the deformed (non-elliptic) confidence regions, can be significantly higher than the computation times of the reparameterization method (assuming that the reparameterization converges). Nevertheless, the gradient-based method provides quite a robust way to calculate the profile likelihood confidence intervals for effective doses. The general positives of the profile likelihood theory were already described in Section 4.3.3. Further, the provided implementation of the gradient-based method proved more robust than the implementation by Dixon et al. (2020).

Higher computation times can be redeemed by the fact that the whole confidence region boundary for parameters (b, e) is found. Therefore, it is enough just to calculate the boundary once to derive all the levels of effective doses required. The effective dose levels can be specified in the `EDx()` call, and the number of required effective dose levels practically does not influence the computation time. Therefore, the gradient-based method can be effective for plotting all the effective doses with corresponding point wise profile likelihood intervals as a function of the dose required (Figure 10) if such a need arises. In comparison, the reparameterization approach would require fitting a new model for each new effective dose of interest.

Further, the calculated boundary of the confidence region can be used to calculate a confidence interval for any surjective transformation of the parameters b and e that does not have local optima, e.g., we could easily calculate a confidence interval for a derived parameter $b+e$, even though such a derived parameter would probably make very little sense.

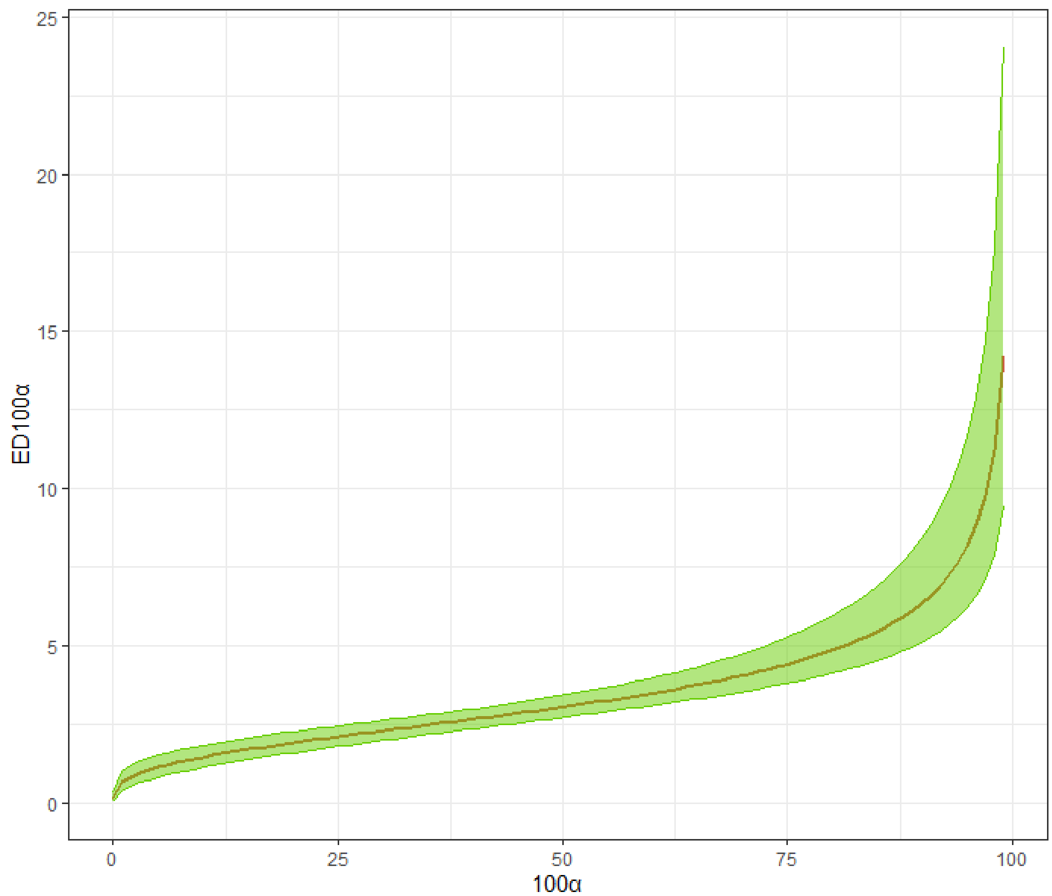


Figure 10: Effective dose as a function of the required level with its pointwise 95% confidence interval for the ryegrass dataset and the LL.4 model.

9 Conclusion

In this thesis a novel approach for finding profile likelihood confidence intervals was proposed. The approach exploits gradient information to find a boundary of a 2-dimensional confidence region that is further used to calculate the confidence interval for derived parameters, without the need for reparameterization. The gradient-based boundary search was implemented for the profile likelihood confidence region of parameters (b, e) of log-logistic and Weibull models with Gaussian, Poisson or binomially distributed response. Subsequent profile likelihood confidence intervals were derived for effective doses for decreasing dose-response curves.

The advantage of the gradient-based approach is its robustness compared to the reparameterization approach. Further, the profile likelihood confidence region for a single fitted model has to be calculated just once and profile likelihood confidence intervals for any/all effective doses can be derived subsequently. The approach is generalizable for derived parameters other than effective doses (with the constraints described in the discussion).

The main drawback is the limitation to a relatively small subset of models and derived parameters. Further, in a small sample settings, a certain degree of manual parameter-setting is needed

Future work should focus on generalizing the gradient-based approach to a wider set of models and derived parameters. Calculating profile confidence intervals (or at least the lower limits of confidence intervals) for benchmark doses in models capturing hormesis could be of further interest. Lastly, a more general method of choosing the tuning parameter λ automatically would potentially benefit the researchers.

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Appendix

A Code

```
loglikelihood = function(object, theta = NULL) {
  if (is.null(theta))
    theta = object$parmMat
  f = object$fct$fct
  df = object$fct$deriv1
  dose = object$data[[object$dataList$names$dName]]
  resp = object$data[[object$dataList$names$orName]]
  weights = object$data[["weights"]]
  N = length(object$data[[object$dataList$names$orName]])
  if (object$type == "continuous") {
    sigma2 = sum((f(dose, t(theta)) - resp) ^ 2) / N
    ll = sum(log(1 / (sqrt(2 * pi * sigma2))) - (1 / (2 * sigma2)) *
               (resp - f(dose, t(theta))) ^ 2)
  } else if (object$type == "Poisson") {
    ll = sum(-f(dose, t(theta)) + resp * log(f(dose, t(theta))) -
               log(factorial(resp)))
  } else if (object$type == "binomial") {
    total = (object$data)[, 5]
    success = total * (object$data)[, 2]
    prob = f(dose, t(theta))
    ll = sum((log(choose(total, success)) +
                resp * log(f(dose, t(
                  theta
                ))) + (1 - resp) * log(1 - f(dose, t(
                  theta
                )))))
    ll = sum(log(choose(total, success)) +
               resp * weights * log(f(dose, t(theta)) / (1 - f(dose, t(
                 theta
               )))) +
               weights * log(1 - f(dose, t(theta))))
  }
  ll
}

score = function(object, theta = NULL) {
```

```r
  if (is.null(theta))
    theta = object$parmMat
  f = object$fct$fct
  df = object$fct$deriv1
  dose = object$data[[object$dataList$names$dName]]
  resp = object$data[[object$dataList$names$orName]]
  weights = object$data[["weights"]]
  N = length(object$data[[object$dataList$names$orName]])
  if (object$type == "continuous") {
    sigma2 = sum((f(dose, t(theta)) - resp) ^ 2) / N
    grad = (1 / sigma2) * colSums((resp - f(dose, t(theta))) *
                                    df(dose, t(theta)))
  } else if (object$type == "Poisson") {
    grad = colSums(((resp / f(dose, t(
      theta
    ))) - 1) * df(dose, t(theta)))
  } else if (object$type == "binomial") {
    grad = colSums(weights * df(dose, t(theta)) *
                     (resp / f(dose, t(theta)) + resp / (1 - f(dose, t(
                       theta
                     ))) - 1 / (1 - f(dose, t(
                       theta
                     )))))
  }
  grad = as.matrix(grad, byrow = TRUE)
  grad[c(1, nrow(grad))]
}


confidence_region <-
  function(object,
           level = 0.95,
           lambda = NULL,
           epsilon = 1e-3,
           maxiter = 2000)
  {
    profile_theta = function(object, point) {
      fixed[c(1, 4)] = point
      fct = fct_gen(fixed)
      if (sum(is.na(fixed)) == 0) {
```

```
    theta = point
  } else {
    temp = update(object, fct = fct)
    theta = fixed[1:4]
    theta[is.na(theta)] = coef(temp)
    theta = theta[is.na(original_fixed)]
  }
  return(theta)
}

descent = function(step = delta, point) {
  beginning = point
  theta = profile_theta(object, point)
  new_loglik = as.vector(-2 * (loglikelihood(object, theta) -
                               loglikelihood(object)) - qchisq(level, 1))
  dff <<-
    rbind(dff, data.frame(
      b = point[1],
      e = point[2],
      ll = abs(new_loglik)
    ))

  descent_counter = 0
  while (!between(new_loglik,-epsilon, epsilon)) {
    grad = score(object, matrix(theta, byrow = TRUE))
    grad = (new_loglik * grad * step) / (norm(grad, type = "2"))
    point = point + grad
    theta = profile_theta(object, point)
    new_loglik = as.vector(-2 * (loglikelihood(object, theta) -
                                 loglikelihood(object)) -
                           qchisq(level, 1))
    dff <<-
      rbind(dff, data.frame(
        b = point[1],
        e = point[2],
        ll = abs(new_loglik)
      ))
    descent_counter = descent_counter + 1
    candidate_counter = candidate_counter + 1
    # print(c(descent_counter, delta))
```

```
    if ((descent_counter == 50) & is.null(lambda)) {
      step = step / 2
      delta <<- delta / 2
      point = beginning
      descent_counter = 0
    }
    if (descent_counter == maxiter) {
      stop("convergence error")
    }
  }

  dff <<-
    rbind(dff, data.frame(
      b = point[1],
      e = point[2],
      ll = abs(new_loglik)
    ))
  df <<-
    rbind(df, data.frame(
      b = point[1],
      e = point[2],
      ll = abs(new_loglik)
    ))
  point
}

generate_rotation_matrix = function(angle) {
  matrix(c(cos(angle),-sin(angle), sin(angle), cos(angle)),
         nrow = 2,
         byrow = TRUE)
}

find_contour = function (rotation) {
  point = beginning
  theta = profile_theta(object, point)
  grad = score(object, matrix(theta, byrow = TRUE))
  grad = (grad * delta) / (norm(grad, type = "2"))
  point = point + rotation %*% grad
  counter = 1
  while (norm(beginning - point, type = "2") >= 0.5 * delta) {
```

```
    theta = profile_theta(object, point)
    grad = score(object, matrix(theta, byrow = TRUE))
    grad = (grad * delta) / (norm(grad, type = "2"))
    point = point + rotation %*% grad
    point = descent(step = delta, point)
    # print(counter)
    # print(point)
    if (counter %% 10 == 0) {
      print(ggplot(df, aes(x = b, y = e)) + geom_point() + theme_bw())
    }
    counter = counter + 1
    if (counter == ceiling(maxiter / 2)) {
      break
    }
  }
  counter
}

start_time = Sys.time()

original_object = object

# Scaling the x-axis for computational purposes
scaling_coef = abs(object$coefficients[["b:(Intercept)"]] /
                     object$coefficients[["e:(Intercept)"]])
object$origData[[object$call[[2]][[3]]]] =
  object$origData[[object$call[[2]][[3]]]] * scaling_coef

if (is(object$fct, "llogistic")) {
  fct_gen = llogistic
} else if (is(object$fct, "Weibull-1")) {
  fct_gen = weibull1
} else {
  stop(paste(object$fct$name, "function not supported", sep = " "))
}

fixed = object$fct$fixed
original_fixed = fixed

object = update(object, data = object$origData)
```

```
# Setting up tuning parameter
if (is.null(lambda)) {
  delta = 1 * min(unname(coef(object)[c("b:(Intercept)", "e:(Intercept)")]))
  delta = level * delta /
    sqrt(length(object$data[[object$dataList$names$orName]]))
} else {
  delta = lambda *
    min(unname(coef(object)[c("b:(Intercept)", "e:(Intercept)")]))
  delta = level * delta /
    sqrt(length(object$data[[object$dataList$names$orName]]))
}

df <- data.frame(matrix(ncol = 3, nrow = 0))
colnames(df) <- c("b", "e", "ll")
dff <- data.frame(matrix(ncol = 3, nrow = 0))
colnames(dff) <- c("b", "e", 'll')
candidate_counter = 0

# Searching for a point on the boundary
beginning = unname(coef(object)[c("b:(Intercept)", "e:(Intercept)")])

if (is.null(lambda)) {
  repeat {
    point = try(descent(step = delta, beginning), silent = TRUE)
    if (inherits(point, "try-error")) {
      delta = delta / 2
      # print(delta)
    } else
      break
  }
} else {
  point = descent(step = delta, beginning)
}

ggplot(dff, aes(x = b, y = e, color = ll)) + geom_point() +
  scale_color_gradient(low = "red", high = "blue")

# Moving alongside the boundary
```

```r
    beginning = point
    runs = find_contour(rotation = generate_rotation_matrix(pi / 2))
    if (runs == ceiling(maxiter / 2)) {
      find_contour(rotation = generate_rotation_matrix(3 * pi / 2))
    }

    if (runs == ceiling(maxiter / 2))
      warning("max_iter reached")

    df$e = df$e / scaling_coef

    list(
      CR = df,
      name = object$fct$name,
      candidates = dff,
      exec_time = Sys.time() - start_time,
      diagnostic_plot = ggplot(dff, aes(
        x = b, y = e, color = ll
      )) + geom_point() + scale_color_gradient(low = "red", high = "blue")
    )
  }

EDfct = function(df, object, perc) {
  if (is(object$fct, "llogistic")) {
    return(exp((1 / df$b) * log((1 - (
      1 - perc / 100
    )) / (1 - perc / 100)) + log(df$e)))
  } else if (is(object$fct, "Weibull-1")) {
    return(exp((1 / df$b) * log(log(1 / (
      1 - perc / 100
    ))) + log(df$e)))
  } else {
    stop(paste(object$fct$name, "function not supported", sep = " "))
  }
}

EDall = function(CR,
                 object,
                 perc,
                 level = 0.95,
```

```
                    lambda = NULL,
                    epsilon = 1e-2) {
  for (p in perc) {
    CR[[paste("ED", p, sep = "")]] = EDfct(CR, object, perc = p)
  }
  CR
}

diag_plot = function(CR) {
  col_names = names(CR$CR)[str_detect(names(CR$CR), "ED")]
  figs = list()
  i = 1
  for (col in col_names) {
    ed_min = CR$CR[which.min(CR$CR[[col]]), ]
    ed_max = CR$CR[which.max(CR$CR[[col]]), ]

    figs[[i]] = ggplot(CR$CR, aes_string(x = "b", y = "e", color = col)) +
      theme_bw() + geom_point() +
      scale_color_gradient(low = "coral3", high = "chartreuse3") +
      geom_point(
        data = ed_min,
        aes_string(x = "b", y = "e", color = col),
        size = 3,
        stroke = 3,
        fill = "white",
        shape = 21
      ) +
      geom_point(
        data = ed_max,
        aes_string(x = "b", y = "e", color = col),
        size = 3,
        stroke = 3,
        fill = "white",
        shape = 21
      ) +
      ggtitle(paste("Diagnostic␣plot␣for␣", col, sep = "")) +
      theme(plot.title = element_text(hjust = 0.5))
    i = i + 1

  }
```

```
  # grid.arrange(figs)
  do.call(grid.arrange, c(figs, list(ncol = length(figs))))
}

EDx = function(object,
               perc = c(50),
               level = 0.95,
               lambda = NULL,
               epsilon = 1e-2,
               maxiter = 2000) {
  CR = confidence_region(object, level, lambda, epsilon, maxiter = maxiter)
  CR$CR = EDall(CR$CR, object, perc, level, lambda, epsilon)
  CI = data.frame()
  lower = paste((1 - level) * 100 / 2, "%")
  upper = paste(100 - (1 - level) * 100 / 2, "%")
  for (p in perc) {
    row = data.frame(paste("ED", p, sep = ""),
                     quiet(ED(object, respLev = p)[, "Estimate"]),
                     min(CR$CR[[paste("ED", p, sep = "")]]),
                     max(CR$CR[[paste("ED", p, sep = "")]]))
    colnames(row) = c("ED", "Estimate", lower, upper)
    CI = rbind(CI, row)
  }
  rownames(CI) <- CI[, 1]
  CI[, 1] <- NULL
  CR$CI = CI
  CR$ED_diag = diag_plot(CR)
  class(CR) = "profileCI"
  CR
}


print.profileCI = function(CR) {
  # print(CR$ED_diag)
  print(CR$CI)
}
```